An Archival ALMA Search for Unresolved Spectral-Carrier Technosignatures toward Stars within 40 pc at 114–873 GHz

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ABSTRACT

We search ALMA archives for narrowband drifting carriers from 61 systems within 40 pc: 405 fine-channel windows, 391 resolving a drift, 47.7 GHz of union bandwidth in 26 disjoint intervals, with 1258 coarser windows constraining only an unresolved excess. Spectra are de-drifted to 12.0–13.3 nHz, 6 per cent of the recommended ceiling; a crossing is one channel and drift rate at 5 times the scatter at control positions. The median system's best window triggers on 1.8×10^{15} W nine times in ten by injection, for a carrier narrower than the instrumental channel. No confirmed carrier is found. Of the crossings, 18 coincide with catalogued molecular transitions, 16 with corroborating evidence: line emission explains the strongest. Unattributed crossings – 31 of the primary census – are consistent with chance against the 24.6 expected at control positions, which does not exclude a smaller added population. No crossing returns to the stellar-frame frequency its own discovery predicts, which is the only recurrence this reading tests. Coverage, not depth, limits confirmation: 21 systems have no repeat at a covered frequency.

Subject headings: Technosignatures, Search for extraterrestrial intelligence, Submillimeter astronomy, Radio spectroscopy, Debris disks

1. INTRODUCTION

The public ALMA archive holds thousands of hours of millimetre and submillimetre observations taken for other reasons, mostly studies of dust discs and planet formation. Their visibilities allow a spectrum at the position of the star at the centre of each field, at a resolution and depth set by the original observer's correlator setup. This paper searches those spectra for a narrowband transmitter at the star's own position. No such signal was found.

Searches for *technosignatures* – detectable imprints of technology, intended or not (Sheikh 2020; Vidal et al. 2026) – have examined many thousands of stars over six decades, from the first proposals and targeted experiments (Cocconi & Morrison 1959; Drake 1961) through Project Phoenix and the programmes reviewed by Tarter (2001) and Backus & Project Phoenix Team (2002) to the present generation of targeted and commensal surveys, several riding back-ends built for other science (Zhang et al. 2020; Czech et al. 2021; Li et al. 2026; Johnson et al. 2023; Choza et al. 2024; Morrison et al. 2023; Tremblay & Tingay 2024; Manunza et al. 2025) and now centred on machine-learning interference rejection (Ma et al. 2023; Jacobson-Bell et al. 2025; Pardo et al. 2025); yet the last published accounting of that effort covers only 6×10^{-18} of the space of transmitter position, power, frequency, bandwidth, polarisation and repetition rate (Wright et al. 2018), and essentially all of it lies below ∼30 GHz. The search used here has no such interference rejection and tests the star instead against control positions within its own primary beam, which a single dish cannot resolve; broadband planetary-scale leakage (Garrett 2026) is outside it by construction.

Above about 30 GHz the record is nearly empty, and that is what makes this frequency range new. Steffes & DeBoer (1994) observed 40 solar-type stars within 23 pc at 203 GHz and Mauersberger et al. (1996) included τ Cet and ϵ Eri among 17 targets at the same frequency, each covering no more than 35 MHz of a single line, with no search over drift rate and no epoch-to-epoch test of persistence. Of ALMA's receiver bands only Band 3 has been searched for technosignatures (Mason et al. 2025): the 35–50 GHz band does hold observations of 4 stars of this sample, but all proprietary until 2027 and all in a continuum mode whose channels are 2048 times coarser than the narrowest channel searched here, so that coverage cannot carry a carrier search, and the band above it holds none at all. Every window reported here was re-reduced and re-searched from the raw archive with the pipeline of §3.

The band has physical advantages and physical costs. Interstellar dispersion and scintillation are negligible at these frequencies, so a carrier arrives essentially undistorted, and a given aperture delivers far more gain per watt supplied. Against that, receiver system temperatures are higher, the atmosphere is opaque in places at both ends of the link, and the same high gain narrows the beam: the 12 m benchmark of §5.3 radiates into a beam 25 arcsec wide at half power, a solid angle of 2.1×10^{-8} sr, so the Earth must lie inside a beam that was deliberately aimed. The limits of this paper are stated as EIRP and need no transmitter model.

Mason et al. (2025) identified the obstacles that shape the present method: drift rates scale with frequency, smearing erodes sensitivity unless drift is searched for, and the molecular line forest raises the confusion background. The quantity that makes the two surveys commensurable is the minimum detectable *received* flux $F_{\rm min} = S_{\rm min}\,\delta\nu$, compared at the channel width each experiment recorded, because $S_{\rm min}$ falls as $\delta\nu^{-1/2}$ while $F_{\rm min}$ rises as $\delta\nu^{1/2}$ and putting a coarse window on a fine channel arithmetically credits it with a depth

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it never reached. On that convention and on native channels throughout, Mason et al. reach 3.8×10^{-23}–5.8×10^{-23} W m^{-2} in a 30.52 kHz channel at a bare 5σ flux threshold, against 1.2×10^{-23} W m^{-2} in a 122.1 kHz channel for the deepest window that lies outside the line mask – the only bandwidth this survey constrains (§3.2) – a factor 4 deeper, and 3.2×10^{-22} W m^{-2} in a 488 kHz channel for the median window, a factor 7 shallower: per channel this survey is the shallower of the two except in its finest windows.

Figure 1(a) places this survey among published searches, in minimum detectable EIRP against distance – a total radiated power and not a spectral power density, for a carrier narrower than every programme's channel, so the axis is commensurable. One non-equivalence governs that comparison and every other in this paper: a limit holds only for a signal narrower than the native channel that set it, and those channels differ by 7 orders of magnitude, from 2.79 Hz to 31.2 MHz, so a signal detectable by one programme need not be detectable by another at any power. Table 1 puts the 3 of 4 programmes drawn that quote another threshold onto a common 5σ basis, which is an arithmetic convention and not a matched false-alarm probability (Appendix D); the filled symbols are those bare thresholds and the open ones the EIRP_{90} quoted everywhere else in this paper, the power recovered nine times in ten through the single-epoch trigger stage and the only symbols on the panel carrying a measured completeness; and no limit drawn in either panel carries a confirmation criterion of any kind. The 24 stars drawn as grey triangles were covered in coarse channels alone, and so constrain an unresolved excess and not a drifting carrier (Table 3). Panel (b) is the other axis, the native channel width of every window searched here beside the channel of each comparison programme: this survey is distinguished by its frequency range and by 61 stellar systems within 40 pc against distant stars in 4 calibrator fields, and not by its depth in any one channel.

The signal sought is an *unresolved spectral carrier*: emission confined to a few channels, unresolved at the stellar position, allowed to drift as it must for a transmitter carried on a rotating, orbiting body (§3.1), and present at more than one epoch. "Narrowband" describes the assumed transmitter and not the resolving power of the archived channels. Recurrence, and not brightness, separates an interesting event from a detection, a signal of interest needs detailed follow-up (Sheikh et al. 2021), and the archive holds a second separated epoch for some of these systems and not for others, so the extent that could be searched and the extent that could be confirmed differ.

A non-detection becomes a quantitative statement only through a stated search volume (Wright et al. 2018), and through its conversion into an upper bound on the fraction of stars hosting a transmitter (Margot et al. 2023; Sheikh et al. 2025) – which is valid only if the sample represents the population it is meant to describe. This one does not: the targets were chosen by other programmes for other purposes, so every statistical statement here is about an archive-selected sample (§2.1), and §2 states what it is instead.

2. THE SAMPLE

TABLE 1
THE COMPARISON PROGRAMMES OF FIGURE 1 (§1). $\mathrm{EIRP}_{\mathrm{min}}$ IS THE DEEPEST LIMIT EACH PAPER REPORTS; THE FACTOR IS 5/(S/N).

programme	S/N	factor	$\Delta\nu_{\mathrm{ch}}$	$\mathrm{EIRP}_{\mathrm{min}}$ (W)	at 5σ (W)
Enriquez + 17	25	0.20	2.8 Hz	5.2×10^{12}	1.04×10^{12}
Margot + 23	10	0.50	3.0 Hz	1.35×10^{13}	6.75×10^{12}
Price + 20	10	0.50	2.8 Hz	2.1×10^{12}	1.05×10^{12}
Mason + 25	5	1.00	30.5 kHz	6.91×10^{17}	6.91×10^{17}

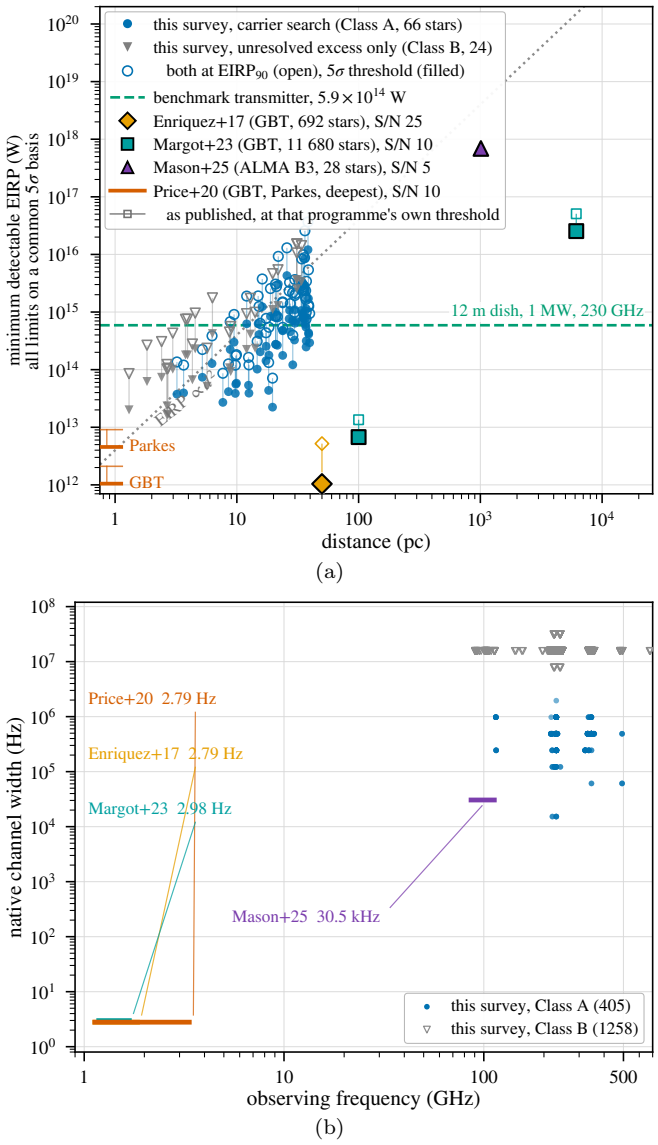


FIG. 1.— Where this survey sits among published radio technosignature searches, on Table 1's common 5σ basis (§1). **(a)** Minimum detectable EIRP against distance; this survey's deepest window per star – blue, the carrier search; grey, coarse channels only; filled, the bare threshold; open, EIRP_{90}. **(b)** Native channel width against sky frequency (filled, Class A; open, Class B).

This search covers 1663 spectral windows toward 90 stars in 83 systems within 40 pc, drawn from 407 public ALMA execution blocks and 4.1 TB of raw data put through one pipeline: 281 h of telescope time, and 1098 h summed over windows. That set is the *primary census*, over which every sensitivity, every expectation and every limit here is computed. It holds 50 of the 56 threshold crossings this paper reports; 4 more are coarse-channel crossings of the same census lying in the one execution block that fails the data-quality criterion of §3.2, reported apart for that reason and not because they fall outside it; and the remaining 2 lie in an *archival tail* of blocks searched after the census was closed, toward stars it already holds, which are not tabulated window by window and enter no expectation. Three further sets enter no census statistic and are reported where their results are (§4.6.1): an *archival extension*, a larger harvest of blocks toward the same stars; a reserved *hold-out*, set aside by the identifier rule of Appendix B.1; and a control beyond 40 pc. Table A1 names every set in rows that sum, and Table A2 resolves every count of windows, crossings and systems into the set it is counted over.

Star means one stellar component, *system* a bound group counted once however many of its components were observed, and *target* one ALMA pointing.

The census is two experiments, divided on native channel width, which is what the search statistic and the completeness calibration both depend on. *Class A* is the 405 fine-channel windows toward 66 stars in 61 systems, of which 391 are channelised finely enough for the searched drift to cross a channel and 14 are not (§3.1); every sensitivity and exclusion quoted here refers to Class A unless stated otherwise. *Class B* is the remaining 1258 coarse-channel windows, which support a secondary search for an unresolved spectral excess. The division is unambiguous because the channelisation of the sample is bimodal: the widest fine channel is 1.95 MHz and the narrowest coarse one 7.81 MHz, a factor of 4 with nothing between them.

Every inclusion criterion was applied by query. A star enters if its Gaia DR3 parallax is at least 25 mas, measured to better than 20 per cent – a clause that never binds, the worst fractional parallax error among the stars searched being 2.5 per cent – and if its proper-motion-corrected position at the observation epoch falls inside the quoted field of view of at least one public execution block. Distances are direct inversions, $d = 1/\varpi$, and the star need not sit at the pointing centre. One further rule applies to the window and not to the star: a window is kept only where the primary-beam response at the star is at least 0.22, the offset at which the Gaussian adopted here and the response of a blocked circular aperture part by 10 per cent on the compact-array dish, the binding one of the two. It removes nothing, in the census or in the archival coverage beyond it, but it does not stand clear of the distribution: the furthest window from a pointing centre lies at $0.82\,\lambda/D$ against a limit at $0.83\,\lambda/D$, and the windows it is nearest are the off-axis ones on ϵ Eridani, which the survey searches. The quoted field of view is optimistic: of the 168 work-list entries the query returns, only 115 stars prove to be contained by a public field, and 90 of those yielded a searchable window. One target has no catalogue name – a serendipitous field star that appears throughout as ALMA J153702653−33192492, Gaia DR3 6015967739554000768, an M5 dwarf at 35.0 pc with no measured systemic velocity.

Disc, planet and colour enter as descriptive metadata only. One exclusion, though, is a property of the catalogue and not of the sky: a Gaia DR3 parallax needs a Gaia DR3 source and the brightest stars have none – every one of the 22 stars in this volume brighter than second magnitude is absent from DR3 astrometry – so 257 of the 4227 Hipparcos stars in this volume whose own parallaxes meet the criteria above never reached the work list – 59 of them brighter than Hp 4 and 198 fainter, a loss saturation does not explain. Of those, seven lie inside a public field of view – Sirius A, α Centauri A, Vega, Fomalhaut, α Centauri B, β Leonis and MCC 124 – and hold 38 fine-channel and 374 coarse-channel windows, 1.0 TB of raw data in 83 execution blocks, none of it searched here, Sirius A among them since the three blocks that put Sirius B in this census are pointed at it. A search of this volume should admit these stars on Hipparcos astrometry where Gaia has none.

2.1. *The archival selection*

This sample was not designed: it is whatever previous ALMA programmes happened to observe among the stars Gaia DR3 catalogues, and 90 per cent of the stars searched were targeted under a debris-disc or planet-formation proposal. Table 2 measures the bias against a volume-limited reference census of which the searched stars are a subset, both columns the same Gaia DR3 selection – $\varpi > 0$, $\sigma_\varpi/\varpi \leq 0.1$ and $1/\varpi \leq 40$ pc, 17 566 objects – and both classified by one rule, in which BP−RP colour sets the class and the absolute G magnitude separates the degenerates that colour alone would place among the A stars, so that every ratio below is like against like. It reaches 17 278 census objects and 88 of the 90 searched stars, the other 2 carrying no Gaia BP or RP photometry; both columns are Gaia-defined, so neither contains the bright stars above and the ratios are internally consistent without them.

Distance is the dominant selection effect: stars within 5 pc are over-represented by a factor 53, and the enhancement falls monotonically with distance to a factor of two *under*-representation beyond 30 pc. Colour comes second – A stars by about 17 and F and G stars by factors of three, because those are the stars with bright debris discs; for the A stars that factor is a floor, the Gaia requirement being itself colour-selective and removing stars as blue as an A star at 3.7 times its rate among all the others. M dwarfs are depleted: 41 of the 90 searched stars, a factor 0.7 of their share of the reference census, and only 28 of them carry fine-channel coverage at all. Coverage is concentrated too, 285 of the 1663 windows coming from 3 systems. No statement in this paper is therefore an occurrence rate for the solar neighbourhood, and the selection acts hardest on exactly the targets a designed search would pick: of the 16 sample systems within 15 pc with a confirmed planet, 6 carry no Class A coverage and so no drift-resolving limit.

2.2. *Frequency coverage and archival scope*

The Class A experiment covers $\Delta\nu_A = 47.7$ GHz of unique sky frequency, and that coverage is sparse: 26

TABLE 2
Selection function of the 90 searched stars against the volume-limited Gaia DR3 census of §2.1. "Ratio": the searched over the census column fraction.

Property	Census	Searched	Ratio
Distance			
$d \leq 5$ pc	48 (0.3%)	13 (14.4%)	52.9
5–10 pc	240 (1.4%)	13 (14.4%)	10.6
10–20 pc	2069 (11.8%)	18 (20.0%)	1.7
20–30 pc	5391 (30.7%)	19 (21.1%)	0.7
30–40 pc	9818 (55.9%)	27 (30.0%)	0.5
Colour class[a]			
A	61 (0.4%)	5 (6.2%)	17.2
F	473 (2.8%)	8 (10.0%)	3.6
G	978 (5.8%)	15 (18.8%)	3.2
K	2221 (13.2%)	11 (13.8%)	1.0
M	13 082 (77.8%)	41 (51.2%)	0.7
Other axes			
Confirmed planet hosts[b]	26 (15.5%)	24 (26.7%)	1.7
White dwarfs[a]	463 (2.7%)	8 (9.1%)	3.4
Searched stars / systems	–	90 / 83	–

[a] Colour bins, not MK types. The boundaries are the colours at which the census's own temperature–colour relation crosses 7500, 6000, 5300 and 3900 K: BP−RP < 0.36 for A, 0.36–0.72 for F, 0.72–0.95 for G, 0.95–1.77 for K and > 1.77 for M. Over the 8594 census objects that also carry a Gaia temperature the two classifications agree for 98 per cent, and every disagreement is between neighbouring bins. A star is counted as degenerate when it lies more than 4 mag below the main sequence measured from the census itself: the two sequences are separated by 5 to 8 mag here, so moving the cut by a magnitude either way changes the count by under 3 per cent. The five classes are normalised over the 80 non-degenerate searched stars and the 16 815 non-degenerate census objects, and the degenerate row over the 88 classified searched stars and the 17 278 classified census objects, which is the same rule in both columns.

[b] Both columns are the same positional join, of the 168-entry ALMA-covered work list and of its searched subset, to the NASA Exoplanet Archive `pscomppars` table accessed 2026 October 4. The 24 searched hosts carry 51 confirmed planets, 22 of the 24 surviving the deuterium-burning mass limit; the Gaia census carries no planet information, so the census column in this row is the work list and not the 17 566-object Gaia census.

disjoint islands between 114.0 and 873.1 GHz instead of a band, so those endpoints bound where observations exist and not what was covered. With the coarse-channel windows the archival coverage is 118.1 GHz in 46 intervals, of which only 114.4 GHz survives the molecular-line mask of §3. Fig. 2 shows the distribution: a transmitter between the islands was never observable here, whatever its power.

The archive is not exhausted either: 177 further blocks lie inside the same query and were unprocessed when the census was frozen, Table A1 accounting for every one of them row by row including the 115 reported here as the archival extension, and the largest single gap is the 18 blocks whose raw visibilities the archive holds but never pipeline-calibrated, 237 GB in all – a gap no effort of ours would close.

3. DATA AND METHODOLOGY

The search looks for one kind of signal: an unresolved carrier at the star's own position, narrow compared with a single correlator channel, and drifting in sky frequency because the transmitter accelerates along the line of sight. Over a long track its power is spread across many channels, so a search that treats each channel on its own will miss it; the analysis therefore *de-drifts* the data before it measures them, shifting each integration's spectrum to follow a trial drift rate and only then adding them together.

3.1. *The de-drifted search statistic*

Each window's dynamic spectrum is stacked along every trial drift rate $\dot{\nu}$. The search statistic is the largest stacked signal-to-noise ratio anywhere on the resulting channel×drift grid,

$$T(\mathbf{x}) = \max_{\dot{\nu},\,\nu} \frac{\sum_t I(\mathbf{x}, t, \nu + \dot{\nu}t) \,/\, \sigma^2(t, \nu + \dot{\nu}t)}{\left[\sum_t 1/\sigma^2(t, \nu + \dot{\nu}t)\right]^{1/2}}, \qquad (1)$$

where $I(\mathbf{x}, t, \nu)$ is the spectrum at sky position $\mathbf{x}$ in integration t and channel ν, and $\sigma(t, \nu)$ is the noise scale there. Both weights are evaluated at the channel the track occupies in that integration, $\nu + \dot{\nu}t$, and not at the channel that labels it. Equation 1 is the inverse-variance-weighted stack of the calibrated intensities along a constant-drift track, divided by the nominal noise of that stack. Three statements about it are kept apart. It is the matched filter for such a carrier, and T an absolute signal-to-noise ratio rather than a local contrast, when the noise is independent between integrations and between channels; that independence is an idealisation and no optimality is claimed where it fails. What the trigger is set against is not that idealisation but the distribution T takes in the data: the 512 control positions give it empirically in every window, on the same visibilities, through the same stack, so any correlation the noise carries is inside them. And what a limit is set against is the injected carriers of §4.1, recovered through the unchanged pipeline, which convert a value of T into a power without passing through the independence assumption at all. In Equation 1 the time t runs from the first integration of the observation, so the frequency ν that labels a track is the one a carrier would occupy at the start of that block, and a repeat epoch is read at the cell the same convention fixes (§4.6). The noise scale is the median absolute deviation across 512 control positions at each integration and each channel, scaled to its Gaussian-equivalent standard deviation: local in time and frequency, global in position, and shared by the star and every control. They lie, from a fixed seed, in an annulus set by each block's own synthesised and primary beams: the field read over is that block's primary beam. The statistic is evaluated at one position at a time, identically at the star and at each control, which is what makes the two comparable. $T_\star$ denotes its value at the star.

Spectra are extracted at the star's apparent position, never from images: for each integration and each native channel the pipeline forms the weighted vector average of the phase-shifted visibilities in total intensity, the mean $I = (XX + YY)/2$ of the two parallel-hand correlations. That average is the dirty-map value at the extraction position – an amplitude, not a deconvolved flux density, so in a field with extended emission it is not a point-source measurement – and the correction for its displacement is measured and not assumed, a carrier injected into the same visibilities returning the normalised dirty beam on that block's own baselines and exceeding the model at the larger displacement, so a limit corrected by it is shallower than it need be (Appendix A.2). The data are used as the archive delivered them, through ALMA's own calibration and quality assurance, with their flagging carried

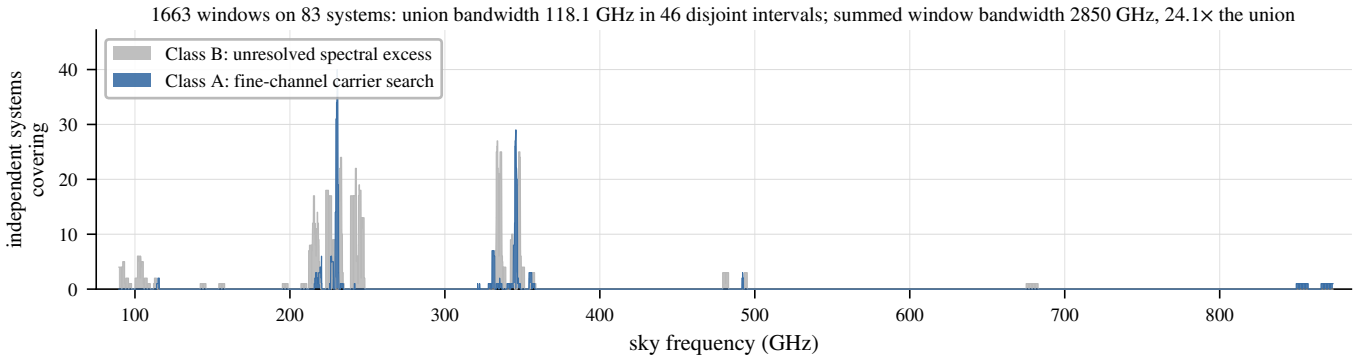


FIG. 2.— Where this survey searched: the number of independent systems covering each sky frequency – blue the fine-channel carrier search (Class A), light grey the coarse-channel spectral-excess search (Class B). Curves step at the searched window edges themselves, so an interval narrower than a plotted pixel still appears. The summed window bandwidth is 24.1× the union.

through, every delivered channel searched, and no regridding, deconvolution or edge trimming; a per-integration median along the *frequency* axis removes the continuum, which removes a level and not a shape, and no median is taken along time, so no feature narrow against the band can be removed by it. One subtlety matters for every power quoted below: the delivered channels are smoothed, so a channel's effective noise bandwidth is not its separation and flux density times channel width is not yet a radiated power, the difference being supplied by a response correction computed from the correlator's own channel shape (Appendix A).

One convention holds everywhere in the paper. The search runs on the topocentric sky frequencies ALMA delivered, and every drift rate quoted is the apparent one at the telescope, which is what the data contain and what a reader can recompute from the archived channel axes. Two steps need the star's own frame and both take it – a crossing is compared with the line catalogue, and a repeat epoch is read at the *predicted cell*, the single channel at a single drift rate where the first epoch's own fit says a carrier should be – by correcting each block's frequency axis for the Earth's motion at that epoch and then for the star's systemic velocity, which is the catalogued radial velocity from SIMBAD (Wenger et al. 2000), queried on 2026 October 4, and zero where the catalogues give none. One star departs: toward β Pic the adopted value is $20\,\mathrm{km\,s^{-1}}$, the stellar radial velocity adopted in the ALMA CO analysis of its disk (Matrà et al. 2017), which this survey's own blocks reproduce and which exceeds the catalogued value by about the template-mismatch offset the *Gaia* pipeline documents for stars this hot (Blomme et al. 2023).

Two power quantities carry the main text, both defined in Table 3: the trigger level $P_{\mathrm{trig}} = 4\pi d^2 S\,\Delta\nu$ at distance d, for a carrier that makes $T_\star$ reach $T_{\mathrm{trig}} = 5$ in one native channel of width $\Delta\nu$, and the sensitivity EIRP_{90}. The second is measured, not assumed: synthetic carriers of known amplitude were injected into calibrated visibilities and recovered through the unmodified pipeline, in three campaigns, one for each stage of the chain the completeness depends on – the trigger completeness itself, on 49 of the 405 fine-channel windows injected directly and on 5 936 carriers of the coarse class; the cost of the spatial rank, on 8 860 carriers over 52 of 56 attempted block–window configurations; and the primary-beam correction, on a third set of 40 scored units – which between them span both ALMA arrays, four bands and the sample's full range of channel widths. Every limit quoted in this paper is the EIRP_{90} of its own window, given to two significant figures beside its native channel width, except the bare-threshold comparisons of §1 and Fig. 1(a), which are labelled as such; the measured completeness is §4.1 and its error budget Table 4. Two comparisons bear on the amplitude scale and they are not the same kind of check: an independent reduction of the same AU Mic visibilities is reproduced to 0.994 ± 0.014, which tests the extraction and not ALMA's flux transfer, an error in which would appear in both reductions alike, while the external comparison with the published quiescent flux density of ϵ Eridani recovers 1.10 ± 0.31 (Burton et al. 2022) and so measures the scale to 28 per cent. The absolute scale is therefore carried at ALMA's own band-dependent accuracy specification: the sample as a whole carries ±11 per cent and the 14 Class A windows in Bands 8 and 10, at 492–493 and 849–873 GHz, carry ±21.

The drift range searched is bounded by an acceleration, not by a frequency. Drift rate and line-of-sight acceleration are one quantity in two sets of units,

$$\frac{\dot\nu}{\nu} = \frac{a_{\mathrm{los}}}{c} \qquad \text{and so} \qquad a_{\mathrm{los}} = \frac{c\,\dot\nu}{\nu}, \tag{2}$$

so a ceiling expressed as $\dot\nu/\nu$ is the same physical limit in every band. The default ceiling is 12.0–$13.3\,\mathrm{Hz\,s^{-1}\,GHz^{-1}}$, the acceleration of a transmitter orbiting a solar-type star at a few hundredths of an au – the Mason et al. (2025) fiducial with a threefold margin, widened for a confirmed exoplanet host whenever the innermost known planet demands more, and 6 per cent of the 200 nHz that Sheikh et al. (2019) recommend for a general search. As an acceleration it is $3.60\,\mathrm{m\,s^{-2}}$, which is also the median over the Class A windows, rising to $4.00\,\mathrm{m\,s^{-2}}$ on the 5 TRAPPIST–1 windows; the grid is uniform in $\dot\nu$, fine enough that a carrier midway between two trials loses no more than 0.26 of a native channel over the longest track, and runs from the negative ceiling to the first trial at or beyond the positive one (Appendix D.6). **The survey does not constrain a transmitter whose apparent acceleration exceeds that range**, and three cases put one there: a deliberately chirped carrier, which is in any case not distinguished from an accelerating platform; a transmitter not bound to a known planet; and the sum of orbital and rotational acceleration, which share the one budget, so a transmitter on a rapidly rotating planet can exceed the ceiling on

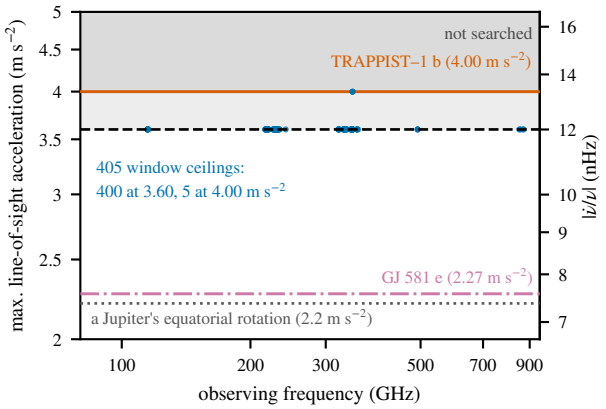


FIG. 3.— What the drift search reaches, as a physical selection function: the drift ceiling of each of the 405 Class A windows (points), by Equation 2, against observing frequency, with the unsearched region shaded and the same quantity as a fractional drift rate on the right. Horizontal lines are the reference accelerations of §3.1.

its spin alone – β Pictoris b, the fastest rotation measured for any planetary-mass object and orbiting one of the stars searched here, corresponds to 19.4 nHz (Sheikh et al. 2019). For the most demanding known case, transiting TRAPPIST-1 b, 29 per cent of the orbital phase lies above the default ceiling, and its maximum line-of-sight acceleration equals the widened grid used on that star's own windows to within the ±3.1 per cent its host's mass and orbit are known to, so the fraction of phase above that grid is anywhere between nothing and a sixth (Fig. 3).

Only part of the archive can carry a drift search at all, and the criterion is whether the maximum searched drift crosses of order one channel or more over the on-source track, $\eta_{\rm drift} = |\dot{\nu}_{\rm max}|\tau_{\rm track}/\Delta\nu_{\rm ch} \gtrsim 1$ – which does not partition the survey the way channel width does: 391 of the 405 Class A windows satisfy it and 14 fall below, in every case because the track is short and not because the channel is coarse, and the released per-window catalogue carries $\eta_{\rm drift}$ for every window and so names them. 38 of the 1258 coarse windows meet the criterion and stay in Class B, because meeting it is necessary for drift discrimination and not sufficient (Table 3); nothing is hidden by that choice, since all of their systems already carry fine-channel coverage and none of the 38 holds a crossing.

3.2. *From a crossing to a confirmed signal*

Every window passes through the same three tests in the same order, and three words name what comes out of them: a *crossing*, an *unattributed crossing* and, had one survived, a *confirmed signal*, each defined once in Table 3 and used in no other sense. There is no fourth test, and none of the three is a significance test. Coincidence with a catalogued transition is a flag and not a disposal, so all 56 crossings continue past it, and the spatial rank and the visibility fit are diagnostics attached to the middle level that decide nothing. Every count reported below is read from the ledger, which lists every crossing the search produced, including those in the archival-tail blocks and in the quality-failing block that no expectation is computed over, so the ledger and the text cannot disagree. The counts are the primary crossings, one to a window: the 21 secondary crossings of §4.3 enter the same three levels, and three reach the second, none the third. The reserved hold-out is reported separately because a transmitter in it would matter whatever its calibration role.

The three tests are defined in Table 3 and are not restated here; what the table cannot carry is the following. The trigger is a level on $T_\star$ and not a detection significance – T is not shown to follow any standard distribution under the null, so the level carries no tail probability, and a window holds between 112 and 6.87×10^6 of those channel×drift cells, so one level buys very different protection in a coarse single-pointing window and in a finely channelised one on a long track. Every cell that reaches the level is recorded, not only the largest in each window, because a carrier is masked as effectively by a noise excursion as by a line. The line attribution is a single database query, made on 2026 October 6 and repeatable as it stands, and its $\pm 50\,{\rm km\,s^{-1}}$ half-width follows the Keplerian line-of-sight velocities of gas surviving beyond a few au with margin for the systemic velocity; a crossing inside it is *attributed* where the species is one that such systems are known to carry, and a further three crossings coincide with a species no debris disc has been shown to carry, whose offsets are reported and which are counted as unattributed (Appendix C). A coincidence is in any case a statement about a catalogue and not a property of a frequency, so this is an attribution and not a disposal, and every crossing goes on to the recurrence test whichever side of the window it falls; the window's price is that the survey constrains nothing inside it. For the recurrence test one reason covers both halves of *separated*: blocks of one session share calibrators, atmosphere, correlator setup and interference environment, and the residual of §4.3.2 is what that can produce, which holds for two blocks of one session however far apart they were executed – 31 pairs of blocks here are of one session and more than a day apart, the widest by 18.02 d. So a recovery in any block of the discovery's own session is a *within-session repeat* whatever the interval, necessary for a confirmation and not sufficient for one; this is also the rule the confirmability census of §5.1 scores the archive on, so the criterion and the scoring are one rule. It is a weak criterion, and the archive's uneven temporal sampling is why: only part of the sample has epochs separated well enough to apply it at all.

Two diagnostics hang off the second test, and the word is exact: each says which crossing deserves attention first, and neither decides anything. The first ranks $T_\star$ against the same statistic at that window's own 512 control positions (§3.1), none of which sits on the star; its lowest value, $1/(512+1)$, means the star beat every control. It is a ranking filter and not a significance test, because the stellar and control positions are not demonstrably exchangeable: the annulus responds to whatever it surrounds, and no parameter-free correction for its geometry restores an exchangeable statistic (Appendix B). The rank distribution does not show that departure, and is quoted with each star carrying equal weight for a measured reason – the five largest stars of the reserved set carry 108 of its 315 windows, so a window-weighted median is a statement about those five: the census median is 0.49 (0.47–0.51) and the reserved blocks' is 0.44 (0.33–0.52), both consistent with the one half exchangeability implies. It also responds to compactness and not to origin, as β Pictoris shows: genuine carbon monoxide ranks below its own control ring there, because the resolved

disc lifts the ring along with the star. The second diagnostic is a visibility-domain fit, which asks whether a crossing's emission is at the stellar coordinates; it cannot discriminate at the trigger either, because at that level a noise maximum at the star fits as a point source there (§4.5). Neither gates a disposition, so neither is charged to the sensitivity. Throughout, p denotes a genuine significance level, and no probability computed against the rank's own null is called one.

One execution block is not a null, and a uniform rule says so. The comparison with the control positions needs each block's ensemble to behave like noise, and the criterion, with the margin by which the one failing block – A002_Xc7fa6f_X28a8 toward BD+05 1668 – fails it, is Appendix D.2's. That block holds 4 of the 56 crossings, which is the survey's entire Class B crossing population, and its windows are all coarse-channel, so the Class A results are identical with it and without it; the quality-passing data therefore hold 52 threshold crossings, all of them Class A. The criterion is applied to blocks, and 1 of those 52 lies in a window the same diagnostic flags inside a passing block; it is counted here and withheld from the amplitude-dependent tests, which is stated where each is made. The 50 in the primary census are the baseline every expectation here is set against and the two in archival-tail blocks enter none of them, while the 4 from the failing block are reported in full in the ledger as the record of a block whose own control ensemble does not satisfy the noise model the rest are compared with, and not as crossings of the same standing (Appendix D).

Because a fixed threshold is not a fixed false-alarm rate, the number of crossings expected by chance is computed window by window from the exceedance rate measured at each window's own control positions; the reserved blocks then test the one assumption that makes, which is that a star behaves like a control position in its own field (§4.3.1). Three corrections precede the statistic: the extraction field is chosen so that the star lies inside the delivered primary-beam field and not merely inside the quoted field of view; the stellar position is propagated to the observation epoch for proper motion, though not for the annual parallax or for orbital motion, whose omissions are corrected window by window in §4.1 and bounded for every component of a multiple in Appendix A.2; and on-source time is accounted per integration from the delivered exposure records and not from the scan intentions. That the chain finds what it is meant to find is shown twice over, by the injected carriers above and by β Pictoris CO – a real feature at a known stellar position, recovered and localised in two transitions and 10 execution blocks (§4.4).

4. RESULTS

The primary experiment is the Class A search for a drifting, unresolved spectral carrier: 405 fine-channel windows toward 61 systems, 391 of which are channelised finely enough on the criterion of §3.1 to tell a drifting carrier from a stationary one. The Class B search for an unresolved spectral excess is secondary and carries no drift discrimination: 1258 coarse windows. Together they are the 1663 windows of the primary census. No event satisfied the confirmation criterion set out in §3.2.

4.1. *Sensitivity: the power at which a carrier would have crossed the threshold*

The median system, on its best window, would have produced a threshold crossing from a carrier of 1.8×10^{15} *W.* A limit is a statement about bandwidth a carrier could have been reported in, so each window's limit is taken over the channels the attribution mask leaves, in the star's own rest frame (§3.2): the finest windows were tuned to CO, a 15.26 kHz window spans 53.9 MHz against a mask 76.9 MHz wide, and 15 of the 405 Class A windows therefore lie wholly inside it and constrain nothing, among them the two finest in the survey. Across the 60 systems left with a constrained window those limits run from 8.7×10^{13} to 3.7×10^{16} W (Fig. 5a), 24 of them reach 10^{15} W and the median Class A *window* reaches 3.1×10^{15} W, while for HIP 10679 every fine window is inside the mask and no limit is quoted. Over the 391 windows in which the searched drift crosses a channel the search covers 44.4 GHz toward 58 systems, and their median system limit is 5.5 per cent deeper than that.

They are a recovery completeness of one epoch, and not a threshold. Artificial carriers were injected at a random sub-channel phase and a random drift from each window's own grid, then recovered through the unchanged pipeline; the power at which nine in ten come back is EIRP_{90}, the only sensitivity quoted here, measured through the single-epoch trigger and the localisation at the stellar position inside one execution block. It is therefore the power at which a carrier would have produced a crossing and not the power at which one would have been confirmed, because confirmation needs a second and separated epoch, which the archive supplies for some systems and not others (§5.1). The 512-position spatial rank decides nothing and is not charged against it either: charging it would cost a further factor 1.31, and in 3 bright-ring windows it could not be met at any injected power (Fig. 4).

Measured that way the completeness is one number for the class. Nine carriers in ten are recovered at $3.03\,P_{\mathrm{trig}}$, with a sampling error of ± 2 per cent on that pooled value – a statement about the class average and about no individual window – against a nominal threshold, P_{trig} being the bare 5σ of the window itself. The loss a channel-confined carrier suffers to the correlator's spectral response is carried by the injected tones, so it sits inside the measured factor, $\times 2.29$ of it, and is not applied to the result as well (Appendix A). Every one of the 49 directly injected Class A windows resolves a ninety per cent point of its own, and the value does not depend on the brightness of its control ring (Fig. 4). Class B carries its own factor $4.17\,P_{\mathrm{trig}}$, which sets the limits for Barnard's Star, Wolf 359 and Proxima, measured the same way from 5 936 carriers injected into 35 coarse windows of 14 execution blocks. Calibration of the class factor is good to $\times 0.89$–$\times 1.11$ (Table 4), an interval on the median and on the counts and not on any one window's limit: the median system's limit above carries it, 1.6×10^{15} to 2.0×10^{15} W, so the second figure of that median is set by the calibration and not by the depth of any observation.

Every limit quoted here is nominal in one respect: it excludes atmospheric decorrelation. An injected carrier is added to already calibrated visibilities and so suffers

TABLE 3
The quantities and the three words this paper uses, defined once. Every term below is used in this sense throughout and in no other. The distinction the first three rows draw is the one the limits of §4.1 turn on: a crossing is a single-epoch event, and a confirmed signal needs a second and separated epoch that no sensitivity supplies.

Term	Definition
Crossing	A channel × drift cell at the stellar position reaching $T_\star \geq 5$ in one execution block; neighbouring channels above the trigger belong to one crossing. The largest in a window is its *primary* crossing and any other a *secondary* one, and because the largest cell is the window's own statistic a secondary crossing exists only where a primary one does. An event requiring investigation; no claimed significance.
Unattributed crossing	A crossing lying more than $50\,\mathrm{km\,s^{-1}}$ in the star's frame from every transition of the line catalogue. That is a reproducible frequency coincidence; which coincidences are astrophysically plausible is a separate judgement, stated with each one. A target for follow-up; not a detection.
Confirmed signal	An *unattributed* crossing recovered at the predicted stellar-frame frequency and drift in a *separated* block: an execution of a different observing session, more than 1 day from the discovery block. A recovery in the discovery's own session is a *within-session repeat*, which a confirmation requires and which does not establish one. The criterion is applied to every crossing, and a crossing coincident with a catalogued transition that satisfies it is a recurring molecular line: 13 of them are, which is this stage's positive control (§4.6). The reading is taken at one channel at one drift, so it holds only for a carrier whose stellar-frame frequency moved by between ±0.08 and $\pm0.63\,\mathrm{km\,s^{-1}}$ at most between the epochs compared. Over the $\pm833\,\mathrm{km\,s^{-1}}$ also searched about that cell, a pair of crossings is tested against a single bounded trajectory wherever the drift grid resolves the interval between them (7 of 13 pairs, Table 5). A pair the test cannot separate is *open*, which is neither a confirmation nor an exclusion. It can be applied only where the archive observed the same frequency again, so for the 21 Class A systems where it did not, no sensitivity makes a confirmed signal possible.
P_{trig}	The power of a carrier that makes $T_\star$ reach the trigger in one native channel. An internal pipeline level: not a sensitivity, not a significance.
EIRP_{90}	The power at which nine injected carriers in ten produce a crossing at the star: a measured single-epoch trigger completeness, not a detection probability. Measured for the 49 injected Class A windows; for the other 356 an estimate from the class-average calibration. Nominal in one respect: it excludes atmospheric decorrelation, which Table F1 gives system by system.
Class A	The primary experiment, defined by channel width: 405 fine-channel windows, median 488.3 kHz, toward 61 systems over 47.7 GHz of union bandwidth, 391 of which satisfy $\eta_{\mathrm{drift}} \gtrsim 1$ (§3.1).
Class B	The secondary experiment: 1258 coarse-channel windows, mode 15.625 MHz, constraining a persistent unresolved excess. Never a carrier search: in 1220 of them no searched drift leaves one channel, and in the other 38 the whole searched range moves a carrier 1.02–1.35 channels, which registers one but does not discriminate it (§3.1).
The sample	Neither volume-limited nor representative of the solar neighbourhood: the stars ALMA happened to observe for other reasons (§2).

none, whereas a real carrier does, and the bias runs one way only. Measured block by block from the phase rms the observatory records on each block's phase calibrator (Cortes et al. 2026, §§10.4.11, 11.3.1) through the standard phase-scatter relation (Richards et al. 2022, Eq. 3), it leaves the median limit optimistic by 1 per cent and the median window above 400 GHz by 8 per cent. *That is why this term is stated and not folded in*: the factors that are applied are available for every window of the census, while this one is available for 279 blocks of 407 and the rest are measurably the harder ones, so folding it in where it is known would leave a set of limits that is neither corrected nor uncorrected. Table F1 gives it system by system (Appendix F).

Three further factors act window by window and are divided out of the limits above rather than quoted beside them – the primary-beam response at the star's own offset from the pointing centre, the annual parallax the search's phase centre omits, and the smearing of a carrier that sweeps within a channel during one integration while an injected one does not – each with its range and its direction in Table 4, its definition in Appendix A or Appendix A.2, and its own value for every window in the released per-window catalogue. *The net bias on one window's limit is therefore of either sign*: in the median window the injected profile's 8 per cent shortfall against the exact channel response is the larger term and the quoted limit is conservative by 7 per cent, while at the ninetieth percentile of the systems' own best windows the decorrelation is, and the limit is optimistic by 18 per cent, rising to 178 per cent for the most affected system.

Carriers were injected directly into 49 of the 405 Class A windows, 12 per cent, and the result transferred to the remaining 356. That transfer is the dominant caveat on everything above. The injected windows are the Class A windows of execution blocks drawn in a random order and balanced by band and array – *the unit is the block and not the window*, because a block's calibrated visibilities are what an injection campaign has to rebuild – and over 6 properties, among them the channel width, the integration time, the noise and the controlling brightness, they span the central eight-tenths of the population, five of the six agreeing in the median to 6.3 per cent and the sixth, the noise level, ×1.7 noisier, which follows from drawing whole blocks (Appendix F). What is carried across is one dimensionless number per class, the ninety per cent point in units of the window's own trigger power, while the channel width, the noise, the integration time, the primary-beam response, the parallax displacement and the intra-integration retention that turn it into a power are each that window's own measurement. Across the injected windows that point ranges from ×0.80 to ×1.58 of the adopted factor, with 41 of the 49 within a tenth of it, and *that dispersion, and not the*

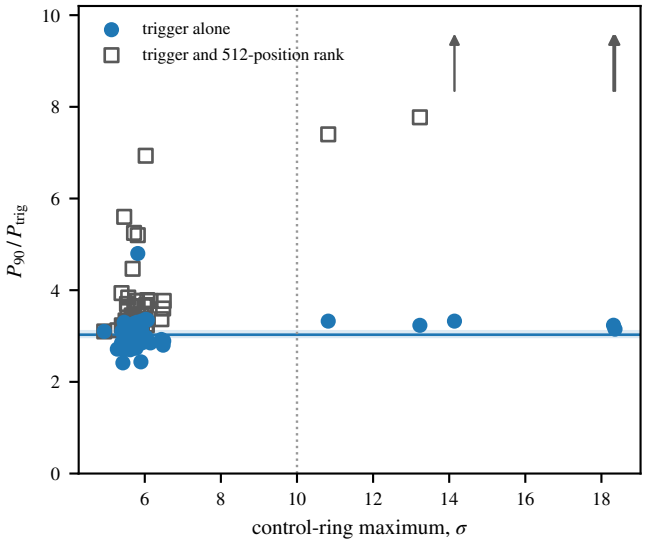


FIG. 4.— Why one completeness factor serves the whole class. The 49 directly injected Class A windows at the power recovering nine carriers in ten, each against its own 512-position ring maximum: no trend with the ring. Filled, the criterion adopted here, pooled value as line and band; open, the rank charged too, arrows where no power meets it; dotted, a resolved ring.

calibration interval above, is what one uninjected window's own limit is uncertain by. Two bands and two channelisations hold no injected window at all, 30 windows between them, and 1 system takes its best window from one – 51 Eri – so in each such case the completeness is carried across a configuration boundary and the range just quoted is not an empirical bound on it. The best window was itself injected into for 6 of the 60 systems, marked as measured in Table F1; for the other 54 the headline limit is an estimate carrying that dispersion.

A carrier that is not always radiating is penalised in proportion: over 3 888 trials at duty cycles down to 10 per cent the recovered amplitude followed the duty cycle to 1 per cent, so a transmitter radiating a tenth of the time needs 10× the power quoted above; those trials carry no drift, so the joint drifting and intermittent case is not measured here.

4.2. *The nearest stars: Barnard's Star and Wolf 359*

Barnard's Star is the second-nearest stellar system and Wolf 359 the third if brown dwarfs are excluded, and to our knowledge these are the first millimetre or submillimetre technosignature limits for either; radio searches of the nearest stars have worked at centimetre wavelengths (Enriquez et al. 2017; Price et al. 2020; Margot et al. 2023). Both are non-detections, and *both limits come from the secondary Class B experiment and not from the carrier search*, the archive holding only coarse Band 6 windows toward these stars: on the EIRP_{90} convention of §4.1 Barnard's Star reaches 2.7×10^{14}–4.0×10^{14} W over 12 windows and Wolf 359 3.1×10^{14}–4.5×10^{14} W over 12.

4.3. *Threshold crossings and how each was disposed of*

The result is a chain of three tests, and it is read in one direction. Of the windows the search covers – the 1663 of the primary census and the archival-tail blocks beside them – 56 hold a cell reaching $T \geq 5$ at the stellar position. 4 lie in the one execution block that fails the quality criterion of §4.3.2, so the population that characterises the search is the 52 in quality-passing data and those 4 are reported beside it. The molecular-line mask is $\pm 50\,\mathrm{km\,s^{-1}}$ wide in each star's own rest frame and places 18 of the 52 within that distance of a catalogued transition; a further three fall on transitions of a species no debris disc has been shown to carry, which the mask does not hold, and stay unattributed with their offsets reported (Appendix C). That leaves 18 crossings attributed and 34 unattributed, and the attributions are kinematic and not an accident of bandwidth: in the frame the correlator delivered, the crossings with a measured peak frequency crowd onto the transitions these setups were tuned to far more tightly than a free landing on their own grids would put them (Appendix E). Of the unattributed, 2 outrank all 512 control positions of their own window, and none was found again in any other epoch covering its frequency. No crossing survives the chain, and this survey reports no detection; the crossing ledger, which accompanies this paper as data, lists every one.

Anything bright in a window hides a weaker carrier behind it, so the weaker cells were sought too. The de-drifted statistic is retained in every channel, so behind a window's tabulated crossing – its largest cell – weaker above-trigger cells can be read as well, neighbouring channels counting as one: there are twenty-one of them, of which three are unattributed and none reaches the trigger in any of the covering blocks the recurrence test reads. Cells are counted in the controls too, and outside the mask the star holds slightly more of them than a control position at every level below the trigger, with no step where a carrier search would put one (Appendix D, §4.3.1).

4.3.1. *The unattributed count is what the control positions predict*

The trigger is a threshold on a statistic and not a false-alarm rate: the number of searched channel × drift cells per window spans 4.8 decades, so one level buys very unequal protection. How many crossings chance should produce is therefore measured window by window, over the 405 Class A windows of the census and with one condition on both sides of the comparison: a cell at the stellar position that reaches the trigger and that the line mask does not attribute. Both quantities are read from the data themselves, and Appendix D gives that accounting. The control positions carry no continuum, so a calibration error varying with frequency would raise the star and not them; on the continuum these stars have, the least demanding window would need 21.4 per cent of it to reach the trigger, and the mild excess of §4.3 runs the wrong way with continuum brightness, 1.01 against 1.06 over the 350 windows with a photospheric estimate (Appendix B). These fields' own control positions predict 24.6 unattributed Class A crossings and 31 are found. A count that large or larger arises by chance with probability 0.13 – a probability conditional on the control positions being a model of the stellar position, which is the one assumption the searched data cannot test. Calibrating the stellar rate against the control rate instead, as a sensitivity test at the ratio 1.104 measured just below the trigger, gives 27.1 expected at $P = 0.26$, and the conclusion of no significant excess is unchanged under

TABLE 4

What sets $EIRP_{90}$, and in which direction. The calibration interval applies to the class-average factor and so to any statistic over many windows; the spread between windows is an observed dispersion and is the statement about one window; the signed terms are biases and are not folded into an error bar (§4.1).

	value	what it does to the limit
Class-average completeness	×3.03	nine carriers in ten recovered at this multiple of a window's own trigger power. *Measured* in the 49 Class A windows injected directly and *estimated*, at this factor, in the other 356
Uncertainty on that calibration (per cent), combined in quadrature		
The factor itself	±2	sampling error of the pooled value over the 49 injected windows
Absolute flux scale	±10.2	the observatory's own band-dependent specification (ALMA Proposer's Guide (Privon et al. 2026), with the Technical Handbook's quality-assurance criterion (Cortes et al. 2026) at Band 8, where the two differ): ±5 in Bands 1–5; ±10 in Bands 6 and 7; ±20 in Bands 8–10, weighted over this sample's bands
Distance and visibility scale	±4.5	the Gaia parallaxes, which enter as d^2, and the block-to-block visibility scale
Combined	$-11/+11$	×0.89–×1.11 on the factor, and so on the median limit and on the system counts; ±21 for the 14 Class A windows in Bands 8 and 10, at 492–493 and 849–873 GHz
Spread between windows. An observed dispersion, not an error bar.		
Window to window	×0.80–×1.58	the 49 injected windows resolve 2.41–4.80 $P_{\rm trig}$ individually, 41 of them within a tenth of the pooled value and the second largest at ×1.11. This, and not the interval above, is what an estimated limit is uncertain by; it is wider, and it applies to 54 of the 60 systems
Applied window by window where a record allows it and bounded where none does, and so not combined with the above.		
Intra-integration smearing	×1.00–×1.05	a carrier sweeps through part of a channel within one integration and the injected ones do not: $\eta_{\rm smear}$ at the drift that sets a ninety per cent point, from each window's own integration length where it is recorded (1054 windows) and from the longest delivered anywhere where it is not (609)
Primary beam	×1.00–×4.3	the star is not at the pointing centre, so each limit is divided by the response at its own offset, on the full width the observatory documents. Nothing in the median window; the 12 windows furthest off-axis carry the departure of that form from a blocked circular aperture, up to 6.2 per cent, as a one-sided systematic
Annual parallax	×1.00–×2.06	the phase centre omits it, so each limit is divided by the amplitude a compact source retains at its own displacement, a median 0.13 per cent on the 1650 windows carrying a parallax record; on the other 13 the loss is bounded (Appendix A.2)
Signed terms. A bias is not an error bar and is not added to one.		
Channel response	×2.29	the share of a channel-confined carrier the peak channel keeps. Carried by the injections, so already inside the completeness above, which is this factor times a margin of ×1.325 over the nominal trigger
its residual	+8	the injected profile falls short of the exact response: *conservative*
Decorrelation	$+0/+1$	measured block by block from the phase rms the observatory records on each block's phase calibrator, which an injected carrier does not suffer and a real one does: +8 above 400 GHz, and more than +20 in 5 per cent of windows

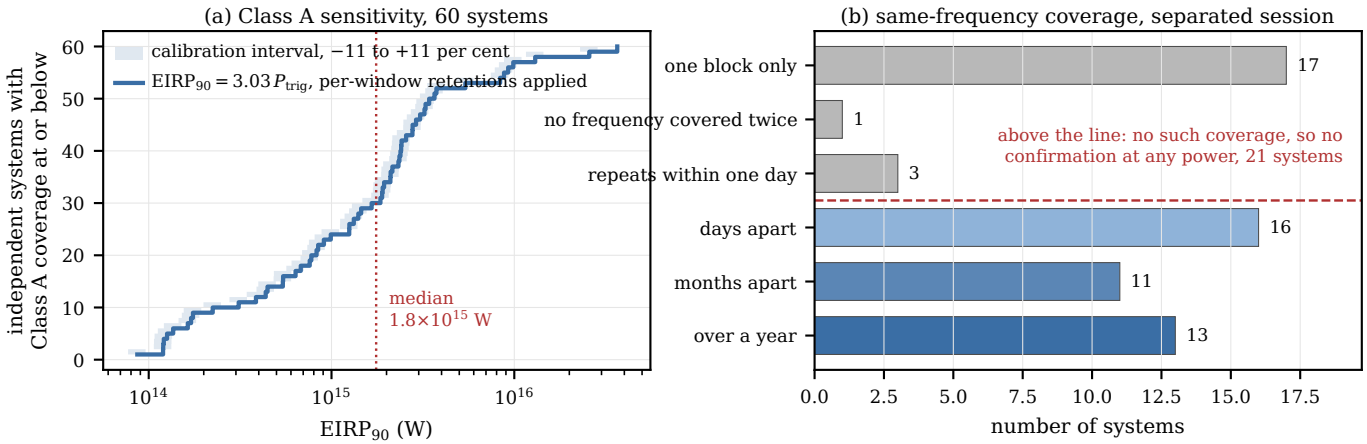


Fig. 5.— **The two completenesses of the carrier search, over the same 61 systems**, which are separate statements and neither of which follows from the other (§5.1). *(a)* Single-epoch trigger completeness: the cumulative number of systems whose best window reaches a given $EIRP_{90}$, each window's own retentions applied and atmospheric decorrelation excluded (Table 4, whose calibration interval is the shaded band). *(b)* Systems with same-frequency coverage in a separated session, which no amount of power buys, grouped by the longest separation between two such sessions. **Left of the dashed line, 21 systems have no such pair, and for them confirmation was not possible at any sensitivity in panel (a).**

it; the control positions' own rate is adopted as the more demanding of the two (Appendix D). Both sides are read over the whole primary beam §3.1 defines, and that is the reading the abstract and §3.2 quote – not because it was nominated first, but because it is the statistic Equation 1 specifies, whereas a narrower field drops the integrations whose pointing falls outside it and with them on-source time and real signal. Read over such a field on both sides the same comparison gives 35 against 24.1 at $P = 0.027$, reported beside the first and not in place of it. Both are extraction geometries applied uniformly to every window and to both sides, so neither was defined by reference to the counts – but neither was nominated as the primary before the counts were compared, and the choice between them was made after the counts had been seen. The whole beam shows no significant excess and the narrower field a nominal one; with 2 readings tested the smaller of the two probabilities, corrected for 2 trials, is 0.054, so neither departs from chance at the 5 per cent level. That correction does not settle the disagreement between them, since it assumes the analysis involved only these 2 choices. The interpretation therefore depends to some extent on the extraction geometry, and neither reading establishes a population of non-astrophysical transmitters. The two disagree on 6 of the 405 windows and in both directions: the whole beam holds a crossing the narrower field does not toward HR 1010 and HD 139084 B, both of them attributed to carbon monoxide, and the narrower field holds one the whole beam does not toward τ Cet, HR 1010 (twice) and η Crv. The stars do carry more above-threshold cells than their own controls below the trigger, and that excess has no identified cause; the rate adopted is the control positions' own, and Appendix D gives both readings and carries the excess as a sensitivity test. A further two unattributed crossings, both toward HD 14055, lie in archival-tail blocks the catalogue does not tabulate window by window, and enter neither side.

The observed count is therefore compatible with a background estimated empirically from these same fields, and that is not the same thing as a null distribution established independently of the model: the attribution applied to both sides, one count per window, and the control positions' own rates. What this comparison could not have rejected is 17 further crossings against those same control positions, 22 on the narrower reading – conditional on that model, a count of crossings and not of transmitting systems, since several crossings could come from one source, not an occurrence rate, and not a statement about technosignatures of any other kind (Appendix D). Each crossing is tested individually by the recurrence test of §4.6.

4.3.2. *One execution block accounts for four of the crossings*

4 crossings belong to BD+05 1668 (GJ 273, Luyten's Star), all of them in one ACA execution block, `A002_Xc7fa6f_X28a8`, whose whole field is above threshold: 512 of its 512 control positions exceed $T = 5$, where the median of a window's own control ensemble is 2.74 across the survey. Its single-integration noise is as predicted but it does not integrate down, a jackknife on the channel mean standing 6.2 times above thermal while neighbouring channels stay uncorrelated, which describes a residual persisting from one integration to the next and differing from channel to channel (Appendix D.2). Equation 1 divides by the noise of the stacked spectrum, which falls as the root of the integration count, so a residual that does not average down accumulates along a drift track faster than the denominator grows, and it does so at every position in the field. *The field is inflated and the star is not*: the 24 sibling executions of the same field, array and setup produce no crossing, and in the strongest of the four more than a tenth of the 512 controls stand above the star. The criterion fails this block and no other at any threshold between it and the next most extreme block, so its 4 crossings are reported apart from the 52 and Appendix E carries what the block might instead be. Milder cases are not unique to it: read window by window the same diagnostic flags 5 windows in 2 blocks, and the 1 flagged crossing outside this block – HD 48370 – is withheld from the amplitude-dependent tests for it too.

4.3.3. *Stellar flares are excluded by a broadband prediction*

Known flare stars are in the sample, but a millimetre flare raises every channel of a window alike, so one able to produce a single-channel crossing at the trigger must produce a $5\sqrt{N}\sigma$ continuum event in the same integrations: with N the window's own channel count, 111–3 769 across the windows holding a crossing, that prediction runs 53–307σ, against an observed continuum statistic of median -0.29 over the 48 crossings with a recoverable measurement and $|Z| < 3$ in 45 of them. A flare origin in fact needs more than that, because the per-integration median along frequency (§3.1) removes a flat continuum before the search: only a flare's departure from flatness across the band can reach a single channel, so the prediction is a floor on what such an origin requires and the contribution measured below is a bound on what one supplies. One block settles it: AU Mic `A002_Xa42f75_X319` holds both a crossing ($T_\star = 5.27$) and the millimetre flare of MacGregor et al. (2020), and that flare contributes 0.17σ to the single-channel statistic – raising the continuum at $Z = 10.3$ while the crossing's own de-drifted track reaches $Z_{\max} = 2.1$.

4.4. *β Pictoris: an astrophysical positive control, and the obstacle it identifies*

A search that finds nothing must show that it could have found something, and the survey's strongest features show exactly that. The pipeline recovers the β Pictoris circumstellar carbon monoxide without being told the disc is there, in two rotational transitions and in 10 execution blocks spanning 8.4 yr drawn from 6 observing sessions, the closest two 25 d apart; in 9 of the blocks the star also outranks every control, but the identification is kinematic and does not rest on those statistics – carried through the frame chain to the barycentre at each block's own epoch and then to the star's rest frame, all 10 crossings lie within $0.5\,\mathrm{km\,s^{-1}}$ of the stellar systemic velocity, in the two lowest CO transitions of a star with mapped circumstellar CO (Dent et al. 2014; Matrà et al. 2017), and the Band 3 peaks agree across blocks to 1.00 channel once each block's own tuning is removed. The alternative reading requires a transmitter placed by coincidence at matched offsets from two CO transitions of one star. It is also the control that shows the recurrence test of §4.6 working in the positive direction: every

flagged feature is recovered in another execution block and each transition in a block of another session as well, the screened Band 6 window returning $T_\star = 8.95$ – and a non-recurrence elsewhere means nothing unless the same machinery recovers a repeat where one exists. *One of those blocks is a counterexample, and it is the more useful half of the control*: in block `X7116f1_X1785`, Band 6, the control ring reaches 14.13 against 9.68 at the star, so several of the 512 control positions exceed the stellar statistic and the spatial rank disfavours a window holding carbon monoxide known to be real and centred on the star. The disc is resolved on the scale of the primary beam in that configuration, which is the geometry the statistic is built to reject, so the rank is a test of compactness and not of authenticity and on a real extended source it errs conservatively – a stronger statement than a clean positive control, because it bounds the diagnostic's reach as well as demonstrating it (Fig. B1). Extended emission is in fact identified by the *level* of the control ring and not by the star's rank against it, which is why the window-level flag of Appendix D.2 is defined on the ring's median, and Appendix C gives the complementary case, HD 48370, whose crossing matches the prominent CO(2→1) that Cataldi et al. (2023) report toward the source and flag as possible cloud contamination, and whose control ring rises with the star as emission filling the primary beam does. The mask that removes this emission covers 11 species and costs 5.0 per cent of the Class A union bandwidth, and it is nevertheless the single most productive test in the chain (Appendix C); §5.1 draws the lesson.

4.5. *The two crossings that lead their own control fields*

Frequencies in this section and the next are in the star's own rest frame, which is the frame the recurrence test registers in; the delivered ones are in Table 6 and in the ledger. Two crossings reach a stellar statistic above every one of the 512 control positions in their own window, and neither is a detection: 61 Vir at 346.3232 GHz and CP−72 2713 at 344.2965 GHz, both in Band 7, tabulated in Table 6 and the first drawn in Fig. 6. Their control rank is recorded because it says which crossing was examined first, and it disposes of nothing (§3). Were the stellar and control positions exchangeable, the star would lead all 512 of them with probability 1/513, so 0.07 of the 34 unattributed crossings would do so by chance and two or more with probability 0.002, against 0 of 8 in the reserved blocks and 0 of 8 beyond 40 pc. That the two observed are unlikely on that model is not evidence of a signal, because the two sets of positions are not exchangeable (§3.2) and neither crossing recurs. CP−72 2713's crossing lies $+1236\,\mathrm{km\,s^{-1}}$ from CS(7–6), a coincidence reported as a note and not as a disposition (Appendix C). Each was re-fitted in the visibility domain, as 51 of the 56 crossings were, by phase rotating to the star and following the recovered drift, and 41 of those fits are consistent with a point source at the star; the fit excludes neither crossing, because injected sources in the bin these two occupy return $\mathrm{Re}/\sigma = 6.58 \pm 1.97$, and its value is that a pedestal fed through a sidelobe from elsewhere in the field – which a dirty map at the star cannot tell from a source at the phase centre – is not a point source there. Of the 512 control positions the rank screen uses, 8 retain a dynamic spectrum and 12 are reachable in the visibilities.

Neither crossing is present when the same stellar-frame frequency is observed again, and Table 6 gives both readings of that silence: the nearest covering block assumes least, lying 1.97 and 2.05 hours from each discovery and ruling out anything above 0.95 and 0.71 of the discovery amplitude, while combining every covering block deepens that to 0.56 and 0.66 but asks the same return over baselines of 3 and 2 days rather than of hours. Either reading covers half a channel in the star's frame, ± 0.211 and $\pm 0.213\,\mathrm{km\,s^{-1}}$, which each carrier's own fitted drift would cross in 77 and 119 seconds: the test asks for a return to that frequency, not a stability at it (§4.6), and Table 6 reports both as the statistic a carrier persisting at the discovery amplitude would have given at the predicted stellar-frame cell, T_{pers}, against the T_{rep} measured there. That reading is one cell at one trial and so may be negative, where each indented reading beneath it is a maximum and so positive on average whatever is present, and the same chance computed at the discovery amplitude itself, which Table 6 gives in brackets, is biased high. At the faintest amplitude consistent with each crossing having been found at all the criterion would have recovered a persisting carrier with calculated probability 1.00 and 0.66, and injected carriers at that amplitude are measured at 0.99 and 0.63 in the accompanying data – so for CP−72 2713 the silence is not an exclusion and is not read as one. Appendix B follows its crossing through every step of the chain.

4.6. *Recurrence, applied to every unattributed crossing*

A transmitter on a body in orbit about its star does not arrive at the same frequency twice: its line-of-sight velocity changes between epochs, and a carrier at a fixed rest frequency changes with it. The question the repeat coverage can answer about such a carrier is therefore the broad one – whether anything above the trigger is present again anywhere in a covering window, searched at every drift of the grid – and it is the result this section leads with. That window is $\pm 833\,\mathrm{km\,s^{-1}}$ wide in the median case, against the $\pm 316\,\mathrm{km\,s^{-1}}$ an orbit at the grid's own $3.6\,\mathrm{m\,s^{-2}}$ ceiling can produce about any star in this sample, whatever its eccentricity and orientation, so it spans the excursion such a carrier could have made. 17 of the 212 distinct covering windows hold a statistic above the trigger, against the 11.3 chance predicts ($P = 0.05$), and every one of those 17 windows holds a crossing the ledger already lists – so this comparison re-reads the trigger-level excess of §4.3.1 over the same control positions rather than adding a recurrence to it. 10 of the 17 are toward HD 14055 and ALMA J153702653-33192492 alone; over the remaining 161 windows the reading is 7 against 4.8 at $P = 0.19$, so such departure as there is sits in two objects and is not a property of the sample. That expectation is a sum of each window's own chance of holding one cell above the trigger, from its independent channel–drift cells and the noise its own control positions measure, and not a control rate applied to their number (Appendix D.5). The strongest of them reaches 5.96 and the nearest sits $56\,\mathrm{km\,s^{-1}}$ from the crossing's own frequency, a median $447\,\mathrm{km\,s^{-1}}$ away. A catalogued window covers 32 of the 33 crossings that carry an amplitude bound and spans that excursion in 27 of them; for 5, toward β Pic, HD 110411 and HD 69830, the widest

covering window is narrower than the excursion, and 1 is covered by no catalogued window at all – CD−57 1054, whose one repeat reading comes from `Xd09670_X3077`, an uncatalogued tail block 38.8 d later. No recurrence is established here, and none is excluded.

A carrier present again would appear in its repeat block as a crossing, so each pair must be asked whether one transmitter could have produced both members. Of the 16 pairs inside that interval with at least one unattributed member, Table 5 asks the question of the 13 across 5 stars whose two crossings each carry an independently fitted drift and a block epoch measured from their own data; the other 3 want a discovery spectrum or a fitted drift (Appendix D.5). Both fitted drifts are carried into the star's frame first, by removing the observatory's own line-of-sight acceleration, which is diurnal here and so cannot be extrapolated across days; Table 5 marks that frame with the subscript $\star$ and carries the stellar-frame offset $\Delta\nu$ beside the two rates, while Appendix D.6 gives the rates in the observatory's frame, the frame the grid is stepped in. Constant line-of-sight acceleration means $\dot{\nu}_1 = \dot{\nu}_2$, which 13 of the 13 pairs fail by more than the grid resolves, and the 7 pairs whose rates the grid resolves fail a drift changing at a constant rate as well. The claim the measurement supports is weaker and safer – that no trajectory inside the searched grid joins the two crossings – and it separates 1 pair, the two crossings of HD 14055 needing 25 times the grid's own ceiling between them; Table 5's *Miss* column is that departure in units of the drift grid's own resolution over Δt, and a dash marks a pair the comparison has no power over at all. The other 12 are unresolved, neither a confirmation nor an exclusion: 10 *open* and 2 *edge-limited*, where one of the two fitted rates sits at the edge of the grid and is a lower bound. The comparison has the power to reject, which it uses on all 7 resolved pairs, and none to accept a drifting carrier – and it does accept the one pair that is a single emitter, the CO(2–1) line of HD 285968, a line that does not drift and which Table 5 carries below its rule as a control row.

A third reading is a check on line attribution, and not a test of persistence. The ledger also reads, for every crossing, the single cell its own discovery predicts – the stellar-frame frequency re-registered at the repeat epoch, maximised over nothing. That cell is the right one for an emitter at rest in the star's own frame, which is what a spectral line is, and the wrong one for the carriers these crossings would be: every one of the 33 has a fitted line-of-sight acceleration of 0.5 to at least $3.6\,\mathrm{m\,s^{-2}}$ in the observatory's frame and at least 3.66 in the star's own, which carries a carrier out of the half-channel the reading covers in 77 to 119 seconds. It is therefore reported here as a *line-recurrence check*, and no crossing is called excluded on it. It works in that direction: 13 of the 15 crossings attributed to circumstellar or interstellar carbon monoxide are present again – 10 in the resolved disc of β Pic and 3 in interstellar gas toward HD 285968 (Appendix C) – each recovered in a block of another observing session more than 1 day from its own, the strongest such recovery 39 to 1984 days away. Close to the trigger the control is thinner than that count suggests: every circumstellar recovery has $T_\star \geq 8.0$, and the 2 attributed crossings not recovered are the two faintest tested, both at $T_\star = 5.41$ and both still consistent with their attribution, so there the check rests on HD 285968 alone. On the 38 unattributed crossings it returns nothing: none is present at its predicted cell, the largest single-block reading at the carried drift being 2.48 against a trigger of 5. What a null there bounds is a carrier whose stellar-frame frequency is unchanged between the two epochs to within half a channel and nothing wider; over the 33 crossings whose repeat coverage carries an amplitude bound that bound reaches a median 0.43 of the discovery amplitude and only 0.997 at the weak end, where 8 of them are too shallow to test at the trigger amplitude (Appendices B.2 and D.4, which also show that no disposition depends on the registration of that cell, on the anchor along the discovery track or on the threshold). Three statements come out of this, and they are not the same statement: no crossing satisfies the confirmation criterion of Table 3; the repeat coverage holds no excess of compatible crossings over chance; and none of the three readings bears on a transmitter whose apparent frequency or acceleration changes between epochs by more than they accommodate.

4.6.1. *The reserved hold-out and the control beyond 40 pc*

Three sets of execution blocks enter no census statistic (Table A1); each was searched with the same pipeline and each is reported here as a search. The *hold-out* is 77 blocks, 315 windows, 36 stars and 214 h on source, set aside by the identifier rule of Appendix B.1 and sharing no execution block with the census; it contains 8 threshold crossings, the strongest at $T_\star = 8.64$, none of them outranking its own control positions, and of the 7 tested for recurrence on the same code path exactly one is present again – a molecular line, so the test works in the positive direction here as well, at $T_{\mathrm{rep}} = +6.71$ – while for every other tested crossing the repeat coverage rules out a carrier at 0.47 of the discovery amplitude or less. The *archival extension*'s 463 windows in 115 blocks hold 13 threshold crossings, every one in a fine-channel window, 5 inside the mask in their own star's frame and 8 outside it against 5.6 predicted by chance (Appendix D), so the chance of 8 or more is 0.20 and a sample this size could have shown a shortfall and does not, $P(0 \mid 5.6) = 0.004$. The *control beyond 40 pc* is 47 blocks and 198 windows toward 22 background sources, 88 h on source, holding 8 crossings, the strongest at $T_\star = 12.7$ and none outranking its own controls; No systemic velocity is catalogued for any of them, so the set supports no limit and shows only that the pipeline behaves the same way on fields beyond the sample. The hold-out crossings are released row by row with the catalogue, on the columns of the crossing ledger.

4.7. *A stellar-frame multi-epoch stacked search*

Combining the epochs of a star in that star's own rest frame deepens the limit. 56 of the 83 surveyed systems hold more than one searched execution block and 59 of the surveyed stars can be stacked, 5 of them from blocks beyond the census; over 320 stacked groups, 2219 epoch-windows and 1431 h the flux-density limit improves on each star's best single epoch by a median factor 1.52 and by 4.45 at most, with 10 further stars, all control stars beyond 40 pc, stacked and excluded from those figures. *A fixed stellar-frame frequency is not a fixed observed frequency, so the epochs are registered before they are added*:

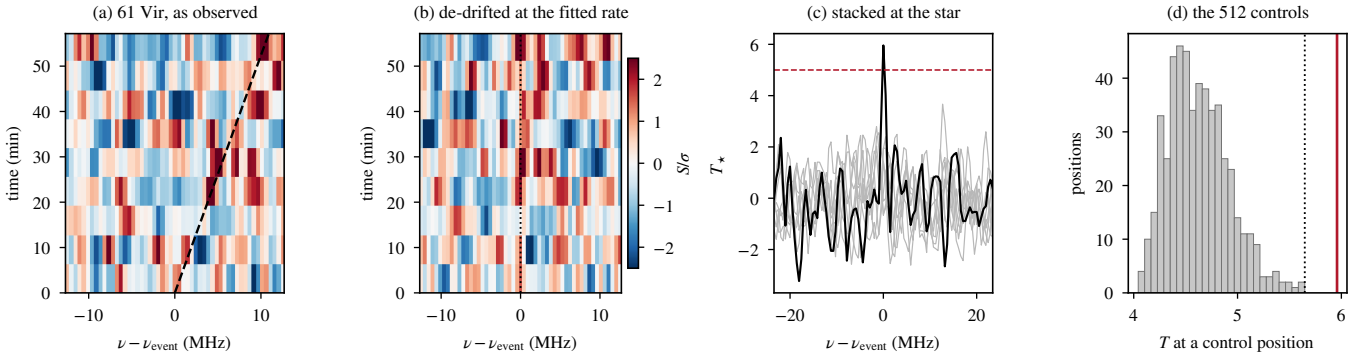


FIG. 6.— **The strongest unattributed crossing of the quality-passing data, and the control field it leads.** Frequencies are in the star's rest frame. *(a)* The dynamic spectrum at 61 Vir's position around the crossing at 346.3232 GHz, observed 2017 May 20, the fitted drift of $+3182\,\mathrm{Hz\,s^{-1}}$ dashed, 225 integrations binned into 9 and plotted in units of each bin's noise. *(b)* The same rows shifted by that drift. *(c)* The de-drifted stack at the star (black) and at the 8 control positions with retained dynamic spectra (grey), in units of its own noise, the trigger dashed. *(d)* The same statistic at each of the 512 rank-screen control positions: the star (solid) against the highest control (dotted).

TABLE 5
CAN TWO CROSSINGS BE ONE CARRIER? THE 13 PAIRS TOWARD ONE STAR WITH AN UNATTRIBUTED MEMBER. *Miss*, THE STARRED RATES, THE CONTROL ROW AND THE VERDICTS: §4.6.

Star	blocks	Δt (d)	$\dot\nu_{1\star}$ ($\mathrm{Hz\,s^{-1}}$)	$\dot\nu_{2\star}$	$\Delta\nu$ (MHz)	miss (σ)	one carrier?
61 Vir	X822/X82f	3.00	+1092	+3221	+755.4	15	open
J1537−3319	X10359/X398	123	−2000	+2390	+223.5	3.6	open
AU Mic	X1838/X319	310	+1440	+2645	−159.0	–	open
HD 14055	X2b2ee/X2c3bb	0.09	−2390	+4216	+825.2	2447	no
	X24be0/Xd46	1.92	+1134	−3823	+63.0	41	open
	X2c3bb/X3835	2.00	+4216	+1800	−514.6	144	edge
	X2b2ee/X3835	2.09	−2390	+1800	+310.6	48	open
	X3835/Xd526	607	+1800	−2563	−555.5	–	open
	X2b2ee/Xd526	609	−2390	−2563	−244.9	–	open
	X3835/X25c5e	649	+1800	+2611	+423.3	–	open
	X2c3bb/X25c5e	651	+4216	+2611	−91.3	–	edge
	X2b2ee/X25c5e	651	−2390	+2611	+733.8	–	open
β Pic	X3a90/Xa9df	1.02	−1611	−679	+85.5	132	open
HD 285968	X108f/X5b4e	4.99	−29	−29	−0.002	0.6	yes

TABLE 6
THE TWO EVENTS, THE CROSSINGS WHOSE STELLAR STATISTIC EXCEEDS ALL 512 CONTROL POSITIONS OF THEIR OWN WINDOW. §4.5 READS THEM.

	61 Vir	CP−72 2713
Frequency, stellar frame (GHz)	346.323236	344.296455
as delivered, topocentric	346.313811	344.269746
Drift rate ($\mathrm{Hz\,s^{-1}}$)	+3182	−2056
$T_\star$	5.958	5.809
Highest control T	5.65	5.68
Fitted amplitude (mJy)	118.8 ± 19.9	11.3 ± 2.0
Peak-channel EIRP (W)	5.1×10^{14}	8.9×10^{14}
Repeat epochs measured	3	5
of them in the primary census	3	1
census repeats with no retained spectrum	1	0
Nearest covering epoch, separation (h)	−1.97	+2.05
covering epochs more than a day away	2	4
Longest baseline (d)	3	2
T expected if persistent, $T_{\rm pers}$	9.70	6.30
T measured, matched drift, $T_{\rm rep}$	+0.39	−0.82
largest single epoch	+1.65	+0.96
maximised over every drift	2.71	3.60
Nearest block alone, $T_{\rm pers}$	5.28	6.21
$T_{\rm rep}$	−0.01	−0.57
Amplitude still bounded, $(T_{\rm rep}+5)/T_{\rm pers}$	0.56 S	0.66 S
from the nearest block alone	0.95 S	0.71 S
from separated blocks alone	0.67 S ($n=2$)	11.79 S ($n=1$)
over the matched registration tolerance	0.67 S	0.84 S
Displacement permitted ($\mathrm{km\,s^{-1}}$)	±0.211	±0.213
Calculated recovery chance Φ at $5S/T_\star$ (at S)	1.00 (1.00)	0.66 (0.90)
Disposition	not confirmed	not confirmed

the barycentric term alone swings by a median $8.2\,\mathrm{km\,s^{-1}}$ between the epochs of one star – many channels at the 488 kHz Class A width – so each epoch is transformed to the stellar rest frame, divided by its own primary-beam response and combined by inverse variance over the channels of common stellar-frame coverage, keyed on the star's catalogued identity; the transform is verified on a real line, HD 285968's, whose 19.26-channel topocentric spread collapses to 0.798 channels in the stellar frame over 4 epochs spanning 44 d. Every stacked limit is measured by injection into the stack itself and never read off a stacked spectrum as a threshold, the criterion is evaluated identically at the stacked control positions so that the whole test's false-alarm rate is measured (Appendix B), and *the gain is* $\sqrt{N_{\rm eff}}$ *and not* $\sqrt{N}$, where $N_{\rm eff} = \sigma_{\rm min}^2 \sum_e \sigma_e^{-2}$ is the equivalent number of equal-depth epochs: the median $N_{\rm eff}/N$ is 0.65 and the realised gain against $\sqrt{N}$ is 0.78, the shortfall being unequal epoch depth, so an epoch count alone misleads.

The deepest stacked limit is Proxima Centauri's, and it is a Class B limit. Proxima is covered by 66 blocks over 85 h spanning 6.3 yr, all of them coarse 15.625 MHz windows, so the stack constrains an unresolved spectral excess and discriminates no drift: stacking the census blocks alone, the set every other limit here uses, gives $\mathrm{EIRP}_{90} = 2.2 \times 10^{13}$ W, and adding the rest of the archive deepens that by 1.26 to 1.73×10^{13} W against 5.51×10^{13} W for the best single epoch on the same convention. *Every stacked limit holds only for an emitter steady in the stellar frame to within one of its own channels, and Proxima Centauri b does not satisfy that.* At 11.185 d about a star of $0.122\,\mathrm{M_\odot}$ – the archive's default parameter set (Akeson et al. 2013), for the planet of Anglada-Escudé et al. (2016) and a host mass agreeing with Kervella et al. (2017) – the planet moves at $47\,\mathrm{km\,s^{-1}}$, so a transmitter on it swings over $94 \sin i\,\mathrm{km\,s^{-1}}$, some 5 channels at $\sin i = 1$, and the stack cancels such a carrier instead of building it unless

the orbit is seen nearly face-on ($\sin i \leq 0.22$, 2.5 per cent of randomly oriented orbits). Within a single block the same transmitter gains at most $1.2\,\mathrm{km\,s^{-1}}$, 6 per cent of one channel, so the single-epoch limit of 5.51×10^{13} W is the one that applies to it – 3.2 times shallower than the stack, and the deepest limit this survey places on the nearest plausible transmitter site. The same tolerance divides the sample: a fine-channel stack admits a far smaller stellar-frame excursion than a coarse one, so the coarse stacks buy reflex tolerance with bandwidth, no crossing of §4.6 drifts slowly enough to survive a fine-channel stack, and the single-epoch limits of §5.3 are the ones that apply generally. *2 groups pass the full criterion and both are known lines*: β Pictoris carbon monoxide in two transitions, the astrophysical positive control of §4.4 passing in the stack as well, and the survey's own $\pm 50\,\mathrm{km\,s^{-1}}$ mask leaves none of them unattributed.

5. DISCUSSION

5.1. *What limits this search*

This experiment's most transferable result is not the non-detection but a measurement of what limits a carrier search at these wavelengths. It is not depth: the power a transmitter must radiate to cross the threshold here is a median 1.8×10^{15} W per system, and more power buys nothing at a frequency the archive never tuned to or never returned to. That inherited coverage sets the limit.

Line emission, and not interference, is the astrophysical false positive here. Catalogued molecular transitions, almost all of them carbon monoxide, account for 18 of the 52 crossings in quality-passing data, and no crossing in this survey is attributed to interference (Appendix E). 16 of those attributions carry support beyond the frequency match – a second transition of the same species, catalogued gas on the sightline, or the crossing's own recurrence – while two rest on the coincidence alone, which leaves them consistent with molecular emission and not identified as it. The attributed crossings include the strongest events in the quality-passing data but are not the commonest: of the 34 unattributed crossings the 31 of the primary census are no more numerous than the control positions of their own fields predict (§4.3.1), so chance at a trigger of 5 accounts for most crossings by number while line emission accounts for the ones that stand out. The remedy that works is a velocity-space mask evaluated in each star's own rest frame, and §4.4 gives what it costs in bandwidth.

Spatially extended emission is the second discriminant, and it has no analogue at centimetre wavelengths. A line filling the primary beam raises the star and the control annulus together instead of cancelling, so the spatial statistic loses power exactly where circumstellar gas is brightest: 10 of the 12 windows that reached the spatial stage are disc or interstellar carbon monoxide, all but one the resolved disc of β Pictoris (§4.4) and the exception a cloud along the sightline to HD 48370 (Appendix C). That one window also dominates every near-edge threshold exceedance in the survey, so the near-edge excess is one interstellar cloud and no edge correction is applied to any limit (Appendix D).

An illustrative engineering benchmark fixes the scale these limits reach: a 12 m dish radiating 1 MW at 230 GHz with an aperture efficiency of 0.7 has a directive gain $G = \eta(\pi D\nu/c)^2 = 5.9 \times 10^8$ and so an EIRP of 5.9×10^{14} W, above the single-epoch crossing limit of 16 of the 60 Class A systems that retain unmasked bandwidth – while the same gain makes its beam $\Omega = 4\pi/G = 2.1 \times 10^{-8}$ sr, illuminating 1.7×10^{-9} of the sky with a half-power width of 25 arcsec, so these limits apply to beamed transmitters whose beam contains the Solar System and EIRP is not transmitter power.

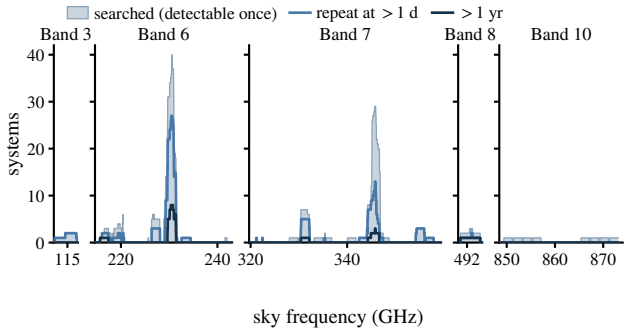


FIG. 7.— What could be searched against where a separated repeat exists. Class A systems against sky frequency: detectable once (grey); with a *separated* block (Table 3) covering it (blue); beyond a year (dark). An accelerating carrier further needs a matching orbital phase. One panel per populated ALMA band, width proportional to its frequency share.

The channelisation is inherited. In 1220 of the 1258 coarse-channel windows every drift in the grid stays inside one channel over a whole track and in 38 more it crosses barely one, so none of them can distinguish a drifting carrier from a stationary one, which is why the carrier search rests on the 405 fine-channel windows. The widths on offer here run from 15.3 kHz to 31.25 MHz, about 3.3 orders of magnitude, which beside the 2.8 Hz channels of centimetre-band carrier searches becomes 7 orders (Fig. 1(b)) – and nothing in the search could choose them.

Repeat epochs turn a crossing into a testable claim, and an archive supplies them only incidentally. Let the *confirmation coverage* $C_{\mathrm{conf}}(\nu)$ be the number of systems toward which two execution blocks of different observing sessions more than a day apart both cover the sky frequency ν, so that a carrier found once there could have been sought again; it must be stated against frequency and not as a fraction of the sample, because repeat visits re-tune and a second epoch is of no use at a frequency it does not cover. Its support is 24.0 GHz of union bandwidth, 50 per cent of the 47.7 GHz searched, falling to 8.5 GHz beyond a year, and at its best near 231 GHz it reaches 27 systems (Fig. 7). A sensitivity limit and a confirmation limit are therefore different statements: this experiment searched 61 systems and 405 Class A windows, and a second session covering a frequency of the first more than a day away exists for 40 of them and more than a year away for 13; for the remaining 21 no such block exists at all, so no transmitted power could have made a confirmation possible. For close to half the sample the archive therefore constrains an intermittent emitter hardly at all, whatever its power.

5.2. *The search space this archive opens*

The one other millimetre search of more than a single line, Mason et al. (2025), covered 117 MHz of Band 3 to-

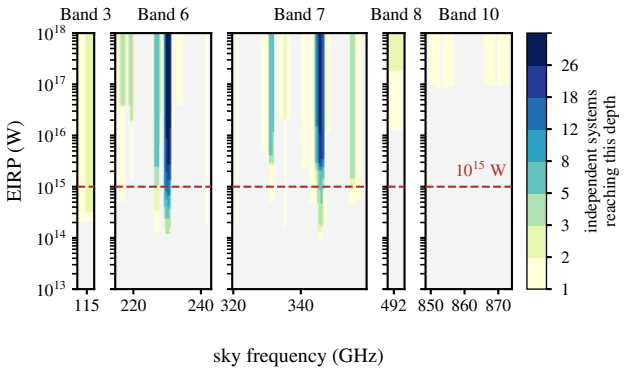


FIG. 8.— Where the experiment could have found something. Shading, the independent systems recovering an unresolved carrier of the plotted EIRP and sky frequency nine times in ten *in a single epoch*; unshaded, none. Confirmation needs a second epoch covering the same cell, which no power supplies. Panels as in Fig. 7, which gives that coverage. Dashed, 1×10^{15} W.

ward far more distant stars (§1); Band 3 is searched here, in 112 windows down to 90 GHz including β Pictoris' CO $J = 1-0$ control.

Union bandwidth overstates what that buys, because different frequencies were searched toward different numbers of stars and to very different limits. Figure 8 gives instead the number of independent systems that would have yielded a recovery at ninety per cent completeness, against sky frequency and transmitter power: at an EIRP of 1×10^{15} W the covered frequency falls to 19.0 GHz, 40 per cent of the union, reaching 24 systems and 16 at the best-covered frequency near 230 GHz, while with no limit on power at all that frequency reaches 38 of the 61 systems and half the sample is reached only above 3.9×10^{15} W.

On the continuous-waveform transmitter figure of merit of Enriquez et al. (2017), with the single survey-wide fractional bandwidth replaced by the star-bandwidth product an archive requires, this experiment is 4.0×10^4 on the median system and 9.5×10^5 on the sample's largest threshold power.[4] That is a factor 2.4×10^5 above the Enriquez value once both are read at the same threshold, which splits into 2.76 decades of haystack and 2.61 decades of depth: a far smaller haystack than a wideband centimetre survey of thousands of stars, and a shallower one.

5.3. *What this experiment excludes, and what it does not*

The signal these limits describe. Every limit here applies to one kind of signal: an unresolved carrier at the stellar position, confined to a single 15.3–1953 kHz channel, lying outside the molecular-line mask, radiating while one of the archival blocks is on sky, and changing frequency linearly at no more than its own window's ceiling, 12.0–13.3 nHz. Such a carrier would have produced a crossing nine times in ten above the nominal EIRP_{90} of §4.1, which leaves out atmospheric decorrelation, and none survived toward any of the 61 systems the carrier search covers. Four conditions bound it as sharply as power does: the coverage is sparse and inherited (Fig. 8); the drift grid reaches an acceleration that contains every known planet of the sample except TRAPPIST–1 b, whose own windows carry a widened grid its maximum equals to the precision its host's mass and orbit are known to (§3.1), while a transmitter on a rapidly rotating planet spends most of that reach on its spin alone (Fig. 3); emission broader than one channel is retained but unmatched by the single-channel filter, which needs close to 32× the total power for a feature as wide as the widest line these fields show, so the thresholds quoted here do not apply to it; and a carrier on for part of a block comes back at that fraction of its amplitude, while one absent throughout is not constrained at all.

The result is a non-detection and not an exclusion of persistent transmitters. Recurrence applies over a far smaller domain than the limits do: where the archive observed a frequency only once no transmitted power makes a confirmation possible, and where the test does apply it has power only against a carrier as strong as the trigger demands of the repeat, so a shallow repeat excludes nothing – 8 of the 33 crossings read at a predicted cell are too shallow to test at the trigger amplitude. A transmitter radiating continuously at a fixed intrinsic frequency is not tested here at all, because its observed frequency follows its own line-of-sight velocity; a return to the same observed stellar-frame frequency is what the reading at a single cell tests, and it is not found; and a transmitter may be intermittent and absent during the second block, may be at a frequency no second block covers over about half the searched bandwidth, may be accelerating as these crossings do and so back inside the half-channel the cell permits only at a matching orbital phase – which is why what bears on such a carrier is the search over the whole covering window and the pair comparison of §4.6, a population statement and not an exclusion of any one of them – or may change frequency deliberately, which nothing here addresses. Outside the adopted criterion, then, nothing is excluded: neither emission inside the $\pm 50\,\mathrm{km\,s^{-1}}$ molecular-line mask at any strength, nor any emission from the stars within 40 pc that ALMA has not observed, which is almost all of the 17 566 this volume holds.

Two blind spots. The analysis forms (XX+YY)/2, which retains the full intensity of a circularly polarised carrier; the loss is not sensitivity but the discrimination polarisation would offer against unpolarised astrophysical emission, because ALMA's feeds are linear and the circularity lies in the cross-hands, which this analysis does not extract. And the frequency coverage is the archive's and not a choice: the 203 GHz positronium hyperfine line that Steffes & DeBoer (1994) and Mauersberger et al. (1996) observed lies in ALMA Band 5, where this survey holds 4 coarse-channel windows, all toward α CMa B, and none of them covers it.

[4] The denominator is the star-bandwidth product $\sum_i \Delta\nu_i/\nu_i$, one term per system, in which $\Delta\nu_i$ is the union of that system's own searched windows and ν_i the midpoint of the span they cover: 0.64 over the 60 systems against 5.8 on a single survey-wide fractional bandwidth, a factor 9.1 apart because the median system was covered over only 1.84 GHz of the 47.7 GHz union. An archive searches different frequencies toward different stars, which a single fractional bandwidth credits to every system alike. The published values are read at the common threshold of Table 1, where 0.85 becomes 0.17 and 0.14 becomes 0.072; Margot et al. (2023) publish no figure of merit, and the value quoted here is the same expression on the 11 680 stars, 1.15–1.73 GHz and $\mathrm{EIRP} > 1.35 \times 10^{13}$ W they do give.

5.4. *What a dedicated ALMA technosignature experiment should do differently*

The changes a purpose-built experiment on the same telescope needs follow from the limitations above.

- *Channelise finely everywhere.* The 1258 coarse-channel windows cannot carry a carrier search at all.
- *Choose the targets and the frequencies.* An archive weighted toward dusty discs is the worst case for molecular-line confusion, the M dwarfs that dominate the nearby habitable-zone planet census are the least covered, and windows placed deliberately away from the bright transitions would recover most of the 2.38 GHz of the 47.7 GHz Class A union that the mask costs.
- *Repeat the epochs on purpose, at the same tuning.* Two or three visits separated by days and by years turn a crossing into a testable claim, and the separation is a scheduling constraint rather than extra sensitivity (Fig. 7). Two epochs cannot test an orbit, only fit one; three separated epochs over-determine it, and 27 of the 61 systems here already carry three at a common frequency, so the shortfall is not coverage but a candidate seen at more than one of them.
- *Retain the visibilities.* Rotating the phase centre to the star leaves a genuine point source entirely in the real part of the visibility, which asks whether the data contain a source where the star is – a stronger question than any rank against neighbouring positions in an image.

6. CONCLUSIONS

1. *What is new.* In the millimetre band the record for nearby stars was two measurements of one spectral line, each no more than 35 MHz around 203 GHz with no drift search and no test of persistence (Steffes & DeBoer 1994; Mauersberger et al. 1996). The carrier search here is 405 fine-channel windows toward 61 systems within 40 pc, 47.7 GHz of unique frequency in 26 disjoint intervals and at least 1363 times that bandwidth, of which 391 windows carry a track long enough for the searched drift to cross a channel; a coarse-channel search for a steady unresolved excess adds 1258 windows toward 82 systems. Both come from one pass through 407 public ALMA execution blocks spanning 90–873 GHz (§2).
2. *How deep.* Taken over the bandwidth the line mask leaves, the median system's best window would have produced a threshold crossing from a carrier of 1.8×10^{15} W nine times in ten, and 24 of the 60 systems with a constrained window reach 10^{15} W; combining each star's epochs in its own rest frame deepens the flux-density limit by a median factor 1.52, the deepest being Proxima Centauri's 1.73×10^{13} W in coarse 15.625 MHz channels, which bounds a steady unresolved excess and not a drifting carrier (§4.7). That is the completeness of a single observation and not of a confirmation, and for 54 of the 60 systems it is carried across from the windows injected into rather than measured in their own. It also leaves out atmospheric decorrelation, which a real carrier suffers and an injected one does not, so the net bias on any one limit runs either way (§4.1).
3. *What was found.* The search produced 56 threshold crossings – 50 Class A in the primary census, two Class A in archival-tail blocks that enter no expectation, and four Class B, all of them from the single execution block that fails the data-quality criterion of Appendix D.2 – so the quality-passing data hold 52 crossings, every one from the carrier search. 18 of those are attributed to a catalogued molecular transition within $\pm 50\,\mathrm{km\,s^{-1}}$ of the star's velocity, circumstellar or interstellar, and 34 are not; an attribution is not an identification, and §5.1 says which carry evidence beyond the coincidence. Of the unattributed, one is atmospheric, leaving 31 of the primary census against the 24.6 these fields' own control positions predict at probability 0.13 – consistency, and not a demonstration that the crossings are noise, since a further 17 would have been consistent too (§4.3.1).
4. *What the repeat observations show.* No crossing satisfies the confirmation criterion. A transmitter on an orbiting body arrives at a frequency its own motion has moved, so the test that bears on one is the broad one, and every unattributed crossing another block covers was searched twice over that way: across the whole covering window at every drift of the grid, where 17 of the 212 distinct covering windows hold a statistic above the trigger against the 11.3 chance predicts ($P = 0.05$), each of them a window whose crossing the ledger already lists; and pair by pair, where of the 13 pairs of crossings carrying enough measurement to be asked the searched grid separates 1 and leaves 12 unresolved, 10 *open* and 2 *edge-limited*. Neither establishes a recurrence and neither excludes one. Nothing returned to the single cell a crossing's own discovery predicts either, the largest single-block reading at the carried drift being 2.48 and 4.71 the largest at any drift of the grid, against a trigger of 5; the same reading recovers 13 of the 15 line-attributed crossings it was tested on, each in a different observing session more than a day away, so the machinery does return a repeat where one exists. What that reading tests is narrow, and the narrowness belongs to the criterion and not to the archive: it requires a carrier radiating in both observations whose stellar-frame frequency comes back to within half a native channel, and a transmitter accelerating at the 0.5 to at least $3.6\,\mathrm{m\,s^{-2}}$ every one of the 33 shows re-enters that interval only if its motion is periodic and the second epoch catches it at the matching phase, so this is not a test of a persistent transmitter of any kind. Of the 61 systems covered, 40 were observed again in another observing session more than a day later at a frequency the first block covered and 13 more than a year later, while for the remaining 21 no repeat exists, so no transmitted power would have made a confirmation possible: coverage is one limit on confirmation and the criterion applied to the repeat is the other (§5.1, §4.6).
5. *What this does not constrain.* These limits hold only inside the domain actually searched: the drift grid stops at 12.0 nHz, 6 per cent of the ceiling recommended for a general carrier search (Sheikh et al. 2019), so an unbound transmitter, one on a rapidly rotating planet, or one whose orbital and rotational

accelerations together exceed that is not constrained here at all (§5.3). The obstacle transfers beyond this archive: channelisation chosen by other observers, and not collecting area, decides which search is possible. Three choices a proposal can simply make would lift it – fine channels everywhere, frequencies and targets chosen against the line forest rather than inherited from programmes that sought it, and repeat visits at one tuning days and years apart (§5.4).

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This paper makes use of the following ALMA data: the 67 projects listed in Table A3, each cited as `ADS/JAO.ALMA#` followed by its project code. The 407 primary-census execution blocks searched span 2013 October 6 to 2025 July 11, and we thank the principal investigators and teams who proposed and obtained them. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), NSTC and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia* (Gaia Collaboration et al. 2023), processed by the *Gaia* Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database (Wenger et al. 2000) (CDS, Strasbourg, France); of the NASA Exoplanet Archive (Akeson et al. 2013), which is operated by the California Institute of Technology under contract with the National Aeronautics and Space Administration under the Exoplanet Exploration Program; and of the Splatalogue spectral-line database (Remijan et al. 2007).

Use of generative artificial intelligence. Large language models were used in this work: they wrote most of the software – the archival retrieval and calibration scripts, the search and vetting pipeline, the statistical tests, and the generators that produce every number, table and figure in this paper. They also drafted text, which the authors revised. The authors set the scientific question, chose every criterion, and are responsible for the content of this paper.

Testing and validation of that analysis. No result here is accepted on a model's assertion: every number in the text is written by the program that computes it from the data rather than transcribed, which removes transcription error and nothing else, and a separate audit rederives the identities that must hold among the published values – counts that must partition, quantities that must sum, numbers reached by two routes that must agree – and refuses the paper where one does not. Every assertion in the analysis was first made to fail on purpose, on deliberately wrong input, before it was relied on, since an assertion that has never fired is no evidence that anything is right.

Software. The pipeline depends on CASA 6.7.6 (CASA Team 2022), PYTHON 3 with ASTROPY (Astropy Collaboration et al. 2022), NUMPY (Harris et al. 2020) and SCIPY (Virtanen et al. 2020), and the ALMA Science Archive and SIMBAD TAP interfaces through ASTROQUERY (Ginsburg et al. 2019); all are free and open-source, at the versions cited.

Author contributions. G.J.W. conceived the survey, defined the target sample and the selection criteria, and wrote the manuscript; R.D. built and operated the archival retrieval, calibration and search pipeline, implemented the spatial control statistic and the injection–recovery campaigns, and produced the per-window catalogue and its regeneration scripts. Both authors designed the vetting procedure, chose the masked species, carried out the statistical validation reported in §3.1 and Appendix D, and agreed the dispositions of the crossing ledger.

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DATA AVAILABILITY

Public. Every observation this paper rests on is public and is identified here precisely enough to be fetched: the 407 primary-census execution blocks, and those of every other set of Table A1, are held in the ALMA Science Archive under the 67 project codes of Table A3 and are retrieved by the execution-block identifier, which the crossing ledger gives for each of the 56 threshold crossings (Chen et al. 2022); the astrometry is Gaia DR3, the planet counts the NASA Exoplanet Archive, and the mask one Splatalogue query of CDMS and JPL made on 2026-10-06 and repeatable as it stands.

Deposited. The machine-readable data behind this paper are deposited as a citable archive (White & Dey 2026): the per-window catalogue of all 1663 searched windows; the Class A and Class B sensitivity measurements; the 56 threshold crossings with their molecular identifications, the atmospheric flag and both field readings of the trigger; the repeat-observation readings; the injection trials and the completeness calibration carried from them; the control-field statistics, including the above-threshold cell counts below the trigger; the execution-block and count accounting, all 21 rows of Table A2 included; and the data behind the figures – every table describing its own columns and units and carrying the project code and execution block that tie each row to an observation, five of them also accompanying the paper on the arXiv as ancillary files. The archive includes short reference implementations of the four definitions the results rest on – the statistic of Equation 1, the control-position generator, the injection–recovery harness and the line mask – runnable against the deposited tables; they implement the published definitions and are not the pipeline that produced the survey.

What the deposit supports. With Table A1 for the block and window partition and Appendix F for every system of the carrier search, it rebuilds the window, star and system counts, the crossing counts and their classification, the chance expectation of 24.6 crossings as a sum over each Class A window's own control exceedance rate, masked share and atmospheric factor, the EIRP_{90} summaries including which windows carry an injected completeness and which a transferred one, and every ex-

clusion depth with its recovery probability. Two quantities it does not give on their own: the interval on the chance expectation, whose resampling draws on quantities that are not per-window, and the completeness of the individually injected windows, whose injection was made into per-integration spectra that are not deposited.

Not public. The calibrated visibilities, the per-window products and the code that produced them are not distributed, so this execution of the search cannot be re-run; the published statistical summaries rebuild from the deposited tables row by row, while whether the description here suffices to implement the search afresh has not been tested.

APPENDIX

A. SPECTRAL RESPONSE, FREQUENCY CONVENTIONS AND THE PRIMARY-BEAM RESPONSE

A.1. *Frequency frames, linewidth and smearing*

Every frequency axis here – searched spectra, tabulated ranges and crossing frequencies alike – is *topocentric* as delivered by the ALMA correlator, and the search applies no velocity-frame shift. Velocities enter at the vetting stage, where the line mask carries every crossing frequency to the barycentre at its own block's epoch and then to the stellar frame before comparing it with a transition (Appendix C). The residual error is then the catalogue radial-velocity uncertainty alone, a few $\mathrm{km\,s^{-1}}$ within 40 pc; second-order relativistic terms at the $50\,\mathrm{km\,s^{-1}}$ scale are $\lesssim 5$ kHz at 350 GHz, below the finest native channel searched. The stacking is drift-aligned and inverse-variance weighted: coherent in frequency trajectory, not in electric-field phase.

The native channel width is also the linewidth convention for every EIRP threshold. The statistic is the single-channel flux density after drift correction, so $\mathrm{EIRP_{trig}}$ is the minimum *total* power of a channel-confined carrier, flat in the assumed linewidth below the channel and never a spectral power density. Above the channel it is not a threshold at all: spreading a fixed total power over N channels lowers the statistic as $N^{0.81}$, measured on Eq. 1 as the power recovered nine times in ten, so a feature as wide as the widest line these fields resolve (69 channels) must carry 32× the power of a carrier, where a filter matched to it would pay 8.5× and this search uses no such filter. The Hz-resolution surveys' limits are total powers in the same sense, each the best case of its own channelisation, which makes the axes of Fig. 1(a) commensurable. Channel widths come from the measurement sets' own CHAN_WIDTH; MS v2 defines no "EFFECTIVE_BANDWIDTH" column, so the quoted channelisations are separations and are not claimed to be noise bandwidths.

ALMA applies online Hanning smoothing by default (Cortes et al. 2026), so the spectral response is ∼2 channels wide and neighbouring channels are correlated. For the 0.25/0.5/0.25 kernel the coefficients are $\rho_1 = 2/3 = 0.667$ and $\rho_2 = 0.167$, and the smoothing narrows the noise by $\sigma_{\mathrm{smoothed}}/\sigma_{\mathrm{raw}} = 0.612$. The peak channel holds 0.50 of an unresolved carrier's flux at a channel centre and 0.42 on a boundary, a median 0.48 over sub-channel phase. *That loss enters the limits exactly once; the thresholds are free of it and the limits are not.* A window's trigger flux is the bare 5σ of its own spectrum, so the *total* power of a channel-confined carrier set against it would be optimistic by ×2.00–×2.36. The injected carriers are not set against it in that way: each is deposited with the correlator's own channel response, at a sub-channel phase drawn afresh, and recovered through the same trigger, which reads the peak channel. The measured completeness therefore already contains the loss, and factorises into it exactly: $3.03\,P_{\mathrm{trig}}$ is ×2.29 for the channel response and ×1.325 for the margin a peak channel needs above the nominal trigger for nine carriers in ten to return. So P_{trig}, the sensitivity S_{min} below and the filled symbols of Fig. 1(a) are nominal throughout, which is what makes them commensurable with other programmes' detection thresholds, while $\mathrm{EIRP_{90}}$ carries the correction once. The deposition falls 8 per cent short of the ideal response averaged over phase, so the completeness is conservative by that much.

Intra-integration smearing, which the injections do not model. Each injected tone is deposited at one frequency per integration, so the campaign carries the channel response and the drift between integrations but not the sweep within one, where η_{smear} is the mean of that response over the sweep and every limit in this paper is divided by it once; the sweep cannot act inside the transform, these channels being 9×10^4 times wider than the reciprocal of an integration, so it redistributes dwell time rather than destroying coherence. The correction exceeds one per cent in 9, reaching ×1.05; it is measured where the integration length is recorded, for 1054 of the 1663 windows, and *bounded* for the other 609 at 11.6 s, the longest integration delivered anywhere, where none falls below 1.000. Re-scoring the 49 injected windows' own ladders with every tone de-rated by its own η_{smear} gives a penalty never larger than the one applied, so the correction errs the safe way.

A.2. *The primary-beam response*

The sensitivity quoted for a window is $S_{\mathrm{min}} = 5\sigma/A$, where A is the primary-beam response at the star's offset from the phase centre: the noise σ is measured in apparent flux density, the limit is quoted in true flux density, and the two differ by A. Archival fields are pointed at a catalogue position, often decades old, while these are high-proper-motion stars, so the star can sit well off axis. Over the 1663 windows the offset has a median of 0.089″ and a maximum of 17.99″; the response A runs from 0.999982 in the median down to 0.2308, falling below 0.9 on 46, every one of those being a J2000 pointing on a fast-moving star rather than an error.

The response applied is a Gaussian of the full width the observatory documents, $1.13\,\lambda/D$ (Cortes et al. 2026), evaluated at each window's *own* dish diameter. The dish matters: the 1058 rows taken on the 1.7× smaller compact-array antennas have a correspondingly wider beam, and a Gaussian forced to $D = 12$ m reproduces the per-window geometry only to 7.7×10^{-2} against 8.7×10^{-5} for the per-dish form. The blocked-aperture response, which carries the shadow of the secondary and is what CASA applies (CASA Team 2022), has a half-power width of $1.15\,\lambda/D$ on a 12 m antenna and $1.14\,\lambda/D$ on a 7 m one; the Gaussian used here is that beam to 2 per cent in width, while a Gaussian of the Airy first-null

width departs from it in response by up to 30.8 per cent, which is why the documented width is the one adopted. The two forms part company as the offset grows, which is what sets the retention rule of §2, and *the furthest window of the primary census lies at* $0.82\,\lambda/D$*, where they agree to 6.2 per cent: that is the largest beam-model uncertainty any sensitivity quoted here carries.*

One star is covered far off axis. ϵ Eridani, at 3.22 pc the nearest star the fine-channel search covers in Band 6, has 20 of its 44 windows at the pointing centre and 24 lying 0.75–0.82 λ/D off it, where the response is 0.23–0.29, inside the main lobe: the blocked-aperture response does not reach its first null until $1.36\,\lambda/D$. 12 of the 44 fall inside the primary census, 8 of them off axis, and carry its statistics, counted in Table A1; the deepest of them is the limit Table F1 gives for this star; no window of it reaches the trigger; and the 32 outside the census, 16 of them off axis, are reported with the archival extension. Nor does any block of this star carry an annual-parallax record, so its limit omits a displacement of $0.22''$ – 0.20 of that block's synthesised beam, from this star's parallax of 310.6 mas, the largest of any the fine-channel search covers – which is smaller than the 0.25 of a beam at which the retained amplitude below was measured, so that measurement bounds the loss instead of being extrapolated to it and the limit is optimistic by at most 9 per cent on this account.

The correction is applied exactly once. The offset is computed for every measurement set from its own `FIELD::PHASE_DIR` and the propagated stellar position, and it enters in one place only: $S_{\min} = 5\sigma/A$ closes on all 1663 catalogued rows to 8.7×10^{-5} in relative terms, and on the 1653 of them whose A comes from the search's own stored geometry to 8.7×10^{-5} – the other 10 having A back-solved from $5\sigma/S_{\min}$ and flagged as such. Nothing else needs correcting: the statistic of Eq. 1, the control ranks and the attribution are *ratios* formed inside one window, in which A divides out identically.

The star's apparent direction includes the annual parallax, and the amplitude a compact source retains when that is left out of the assumed phase centre is measured rather than modelled. Re-extracting the 8 windows of 2 blocks furthest out at the full apparent position – proper motion, parallax and radial velocity where catalogued – leaves each window's noise unchanged to 0.20 per cent and alters no disposition. A carrier injected at the apparent position and recovered from the *same* visibilities at both does return the normalised dirty beam, with no model in it: 0.92 against a predicted 0.89 at 0.25 of a beam, and 0.58 against 0.49 at 0.67. The measurement exceeds the model at the larger displacement, so a limit corrected by the model is shallower than it need be.

A component of a bound pair does not travel in a straight line, and the propagation assumes it does. Of the 91 searched positions, 30 have a companion inside 400 arcsec (Mason et al. 2001; Matson et al. 2026) and 13 more have a disturbed astrometric solution with no companion found. Where the pair has a published visual orbit the displacement follows from it, where it has not the projected separation puts a ceiling on the orbital acceleration with the companion's mass taken from Pecaut & Mamajek (2013), and the Hipparcos–Gaia proper-motion anomaly (Kervella et al. 2022) covers 23 of them. The largest displacement anywhere is 0.20 arcsec at UCAC2 20312880, 0.28 of that block's synthesised beam, no crossing lies more than 0.007 of a beam from its predicted position and the deepest limit of any system moves by at most 16 per cent – except for Barta 161 12, HD 155555C and LHS 1140, whose solutions are disturbed with no companion located and no Hipparcos epoch to measure one from, so that the 24 windows of those 3 stars carry limits optimistic by an unmeasured amount.

Table A1 accounts for every execution block the archival query exposed and Table A2 for every event the search returned; the full itemisation of both, all 21 rows, accompanies this paper as data, and the row numbers below are that itemisation's. The telescope on-source time behind the primary census is 281 h, of which 153 h carries at least one Class A window and 277 h at least one Class B window – figures that overlap and must not be added. Confirmed planet counts there and in the target list come from the NASA Exoplanet Archive `pscomppars` table (Akeson et al. 2013), accessed 2026 October 4. Three counts in the second table are 52 over three different sets – the Class A crossings of the adopted ledger, those of the per-window extraction catalogue and the quality-passing crossings – and the first two differ on 6 rows, where the statistic read over a narrower field falls on the other side of the trigger from the whole-primary-beam reading the ledger adopts, in both directions (Appendix D); the accompanying table carries the ledger's flag, 50 of them in the census. Row 3.1.1 applies the criterion of §3.2 and row 2.2.1 is decomposed system by system in Appendix D.

B. THE SPATIAL RANK, AND WHAT IT CAN SUPPORT

The spatial rank asks whether an excess sits at the star or anywhere else in the field. It is a diagnostic, no disposition rests on it, and this appendix is the measurement that settles why: The star-versus-control rank is an empirically useful way of deciding which crossing to examine first, but the stellar and control positions are not demonstrably exchangeable under the null. The rank therefore has no calibrated per-window false-alarm probability and is not used as evidence for detection.

The star lies at the phase centre, inside the control annulus, and the displacement is real: the star carries more above-threshold cells than a control position at every level below the trigger. The chance expectation of §4.3.1 is not stepped by it, since the ratio at the trigger is formed from the cells under test, and what carrying it in would do is measured in Appendix D. *It is not a primary-beam effect.* Spectra are extracted from phase-rotated visibilities in apparent flux density and are never corrected for the primary-beam response (Appendix A), so the familiar rise of image noise with distance from the pointing centre cannot operate here; were it operating it would make the control probes noisier than the star and place the stellar rank *above* one half, the opposite of what is measured. The control ensemble does show a weak rise of its own level with radius, and each probe is therefore standardised within its own radius bin, with no free parameter.

A calibration error that varied with frequency would

TABLE A1

The survey ledger, part one: execution blocks and windows. Each row carries its own number, and a group sums to the row whose number heads it. Hours are window-hours, so the classes partition the hours as they partition the windows. Rows 3.2 to 3.5 are the archival gap, over which no result is formed.

	Set of execution blocks	Blocks	Windows	Stars	Systems	Hours
1	Archival scope: blocks the query exposes	656	–	–	–	–
1.1	… processed with the same pipeline	479	–	–	–	–
1.2	… not processed at the freeze	177	–	–	–	–
2	Processed with the same pipeline[c]	484	–	–	–	–
2.1	**Primary census**	407	1663	90	83	1098
2.1.1	Class A, fine channels[a]	–	405	66	61	228
2.1.2	Class B, coarse channels[a]	–	1258	89	82	870
2.2	Reserved hold-out	77	315	36	–	214
3	Not processed at the freeze: what became of each	177	–	–	–	–
3.1	Calibrated and searched	144	–	–	–	–
3.1.1	**Archival extension**, reported	115	463	16	16	269
3.1.1.1	Class A, fine channels[a]	–	67	11	–	48
3.1.2	Searched after the extension was frozen	8	32	–	–	–
3.1.3	ϵ Eri, both pointings[d]	8	32	–	–	–
3.1.4	No window reached the export	13	0	–	–	–
3.2	No calibration in the ALMA delivery	18	–	–	–	–
3.3	Calibration failed	11	–	–	–	–
3.4	Not attempted	3	–	–	–	–
3.5	Working disk insufficient	1	–	–	–	–
4	Beyond 40 pc: out-of-sample control	47	198	22	–	88
5	Rank-calibration sample[b]	326	1322	56	–	–

[a] The two classes partition the census windows, not its blocks: 255 blocks carry a Class A window and 392 a Class B one, and most carry both, so the hours of the two rows partition the windows and not the observing time.

[b] Not an out-of-sample set and not an addition to this ledger: every one of its blocks is counted in a row above, in the primary census or in the hold-out, and it is listed apart only to keep it from being counted twice.

[c] Panels 1 and 2 overlap rather than add: 479 of these 484 blocks are row 1.1, and the other 5 carry no member observing unit in the archive metadata, so the scope query does not expose them; 4 of the 5 are in the primary census.

[d] Every one is an ϵ Eri Band 6 block. The archival pointings put the star either at the centre of the field, at a response of 1.00, or well off it, where it is 0.23–0.28 over the 16 of this row's 32 windows that lie off axis, above the 0.22 retention floor of §2 in every case. All of them are searched, each from its own recorded phase centre, and their conventions are in Appendix A.
The same band's 12 windows that fall in the primary census are counted in row 2.1 and carry every census statistic; the on-axis pointings reach a response of 1.00.

explain the displacement, and it is the only mechanism on offer. An error identical in every channel is removed exactly by the per-integration median along frequency; a channel-dependent one is not, and two ordinary errors are channel-dependent: a residual in the bandpass, and a system temperature measured in the coarse correlator mode and interpolated onto the science channels, which near a telluric line leaves a narrow feature behind (Hunter et al. 2018). Both multiply the continuum, so both act at the star and not on empty sky, both stand still in topocentric frequency and both are coherent between integrations: they would stack at near-zero drift and put the stellar rank *below* one half, the sign observed. *Amplitude excludes them, measured where the search looks.* Such an error multiplies the continuum at the extraction position and must beat the noise of the de-drifted, block-combined spectrum, since a coherent residual stacks as a carrier does, so a fractional error ϵ depositing ϵS_c at the star needs $\epsilon = 5\sigma/S_c$ to reach the trigger. That continuum is measured for each window in its own execution block, as the time average of the window's own retained spectra, over 1557 of the 1663 windows, and over 214 blocks it agrees with independent continuum images of the same fields with no offset between the two (-0.001 mJy). No window requires as little as the 5–20 per cent a coarse system temperature is documented to leave at an ozone line – the smallest requirement anywhere is 21.4 per cent and the median 1067, and over the 51 crossings with a measured continuum the least demanding needs 53 per cent – while the bandpass branch fails by amplitude everywhere, needing 21× the 1 per cent variation the observatory quotes for its worst bands (ALMA Science Portal 2026; Kameno 2014); and of either sample 1 crossing lies within 62.5 MHz, two of the 31.25 MHz channels on which the system temperature is measured, of a catalogued atmospheric transition (Appendix E). The shared noise scale raises one further question, and the same arithmetic settles it: over the 1609 of the 1663 census windows holding none of the adopted crossings, the stellar statistic divided by the median of the 512 control statistics is exactly the ratio of the two noises, its median 1.001 (0.994–1.005), so the two scales agree.

B.1. *The reserved hold-out, and what it calibrates*

The statistic and the mask are set on the searched sample and cannot also calibrate themselves, so a further sample is separated by an identifier rule: a block is held out if and only if $\mathrm{sha256}(\mathrm{UID})_{[:8]} \bmod 5 = 0$, subject to two guards that use identifiers alone – a block on a named list stays in the survey, and no system loses every block. That this sample is independent of the choice of statistic and of mask is an assertion of ours and not something a reader can check. Of 484 blocks processed, 77 are reserved, 15.9 per cent of them, carrying 315 windows toward 36 stars, with no star and no system lost from the headline sample. The rule hashes the execution block, and ALMA's identifier carries the observing session above it, so 64 per cent of the reserved set – 49 of the 77 blocks, across 40 sessions – shares calibrators, weather and correlator setup with a searched block, where a session-level hash would have been the stronger choice; every quantity drawn from this sample is therefore given twice, over the whole reserved set and over the 28 blocks that share no session with it. One further quantity is calibrated on those blocks and on nothing else, the

TABLE A2
The survey ledger, part two: events, dispositions and confirmation. Every headline count in this paper is read from here.

	Count	Value	What it counts
1	Threshold crossings at $T_\star \geq 5$	56	census and archival tail
1.1	Class A, fine channels	52	the carrier experiment
1.2	Class B, coarse channels	4	all in the one block failing §4.3.2
2	Quality-passing crossings	52	the same rows as 1.1
2.1	… attributed to a transition	18	of 18 coincidences inside the mask in the star's own frame
2.2	… unattributed	34	rows 2.2.1 to 2.2.3
2.2.1	… in the primary census	31	against 24.6 expected by chance at the 405 windows' own control positions – a count chance delivers with probability 0.13
2.2.2	… in archival-tail blocks	2	outside that comparison
2.2.3	… coincident with an atmospheric transition	1	within 62.5 MHz of one in the topocentric frame (Appendix E), and counted as atmospheric on both sides of that comparison
3	Class A systems	61	of 83 surveyed
3.1.1	Class A systems with same-frequency coverage in a separated session	**40**	**the only confirmation count quoted**: a later block in another session covering a frequency the first covered, 13 of them over a year later. For an accelerating carrier, confirmation further requires the repeat to fall at a matching orbital phase. Over the census together with the repeat blocks the recurrence test reads
3.3	… with no separated second block at a covered frequency	21	confirmation impossible over the 166 blocks the recurrence test reads; 36 of all 83
4	Class A union bandwidth	47.7 GHz	26 disjoint islands
4.1	… removed by the line mask	5.0 %	2.38 GHz: the figure the text quotes, against 3.1 per cent of the 118.1 GHz archival union

TABLE A3
The 67 ALMA proposals the searched blocks belong to, as ADS/JAO.ALMA#*code*. Every census block belongs to one.

2012.1.00142.S, 2012.1.00198.S, 2012.1.00268.S, 2012.1.00385.S, 2012.1.00437.S, 2012.1.00993.S, 2013.1.00359.S, 2013.1.00588.S, 2013.1.00645.S, 2013.1.01147.S, 2013.1.01214.S, 2015.1.00307.S, 2015.1.00633.S, 2015.1.00783.S, 2015.1.01015.S, 2015.1.01260.S, 2016.1.00104.S, 2016.1.00358.S, 2016.1.00515.S, 2016.1.00637.S, 2016.1.00803.S, 2016.1.00817.S, 2016.1.00878.S, 2016.1.00880.S, 2016.2.00015.S, 2016.2.00200.S, 2016.A.00013.S, 2017.1.00167.S, 2017.1.00200.S, 2017.1.00215.S, 2017.1.00561.S, 2017.1.00698.S, 2017.1.00704.S, 2017.1.00786.S, 2017.1.00986.S, 2017.1.01583.S, 2017.1.01644.S, 2018.1.00045.S, 2018.1.00072.S, 2018.1.00126.S, 2018.1.00461.S, 2018.1.00470.S, 2018.1.00947.S, 2018.1.01149.S, 2018.1.01545.S, 2018.1.01833.S, 2019.1.00189.S, 2019.1.00696.S, 2019.1.01443.T, 2019.1.01517.S, 2019.1.01603.S, 2019.1.01681.S, 2019.2.00208.S, 2021.1.00485.S, 2021.1.01209.S, 2021.2.00097.S, 2022.1.00134.S, 2022.1.00338.L, 2022.1.00793.S, 2022.1.01163.S, 2023.1.00390.S, 2023.1.00487.S, 2024.1.00165.S, 2024.1.00595.S, 2024.1.00722.S, 2024.1.01095.S, 2024.A.00040.S

radial behaviour of the control ensemble: the standardised level runs +0.12, +0.10, +0.12, +0.16 across the annulus, a mean per-window slope of $+0.067 \pm 0.033$, small and of the wrong sign to account for the displacement.

B.2. *The recurrence statistic under the null*

The bound in the crossing ledger is one formula. Writing S and σ_d for the discovery amplitude and its error, and $\bar{a}$ and $\bar{\sigma} = (\sum_i \sigma_i^{-2})^{-1/2}$ for the inverse-variance combination of the repeat readings at the predicted channel,

$$\sigma_\mathrm{excl} = \frac{S - \bar{a}}{\sqrt{\sigma_\mathrm{d}^2 + \bar{\sigma}^2}} = \frac{T_\mathrm{pers} - T_\mathrm{rep}}{\sqrt{1 + (T_\mathrm{pers}/T_\star)^2}}, \qquad \text{(B1)}$$

with $T_\mathrm{pers} = S/\bar{\sigma}$ and $T_\mathrm{rep} = \bar{a}/\bar{\sigma}$, so that every bound follows from Eq. B1 and three columns of that table and nothing else: T_rep is read at the carried drift and at one cell, and the largest value over every drift trial at the same cell is tabulated beside it but is never compared with T_pers, since a maximum is positive on average whatever is there; an amplitude selected as the largest of 10^2 to 10^7 cells at $T_\star \simeq 5$ is biased high, so the limit $(T_\mathrm{rep} + 5)/T_\mathrm{pers}$ is the form of the statement that survives the bias; the reading itself is centred on zero and scatters by 1.046 about a mean of +0.014, measured over 511 single-cell readings of every repeat block in the census and the hold-out; where the bound has power is $\Phi(T_\mathrm{pers} - 5)$ for a carrier radiating through both observations at the discovery amplitude and $\Phi(5T_\mathrm{pers}/T_\star - 5)$ at the trigger amplitude $5S/T_\star$, the faintest that could have produced the crossing at all, so that where they are small the crossing is untested and the 8 smallest are reported that way; and each crossing's own recovery chance, calculated and measured by injecting a carrier at the trigger amplitude at its predicted cell through the instrument's own channel response, accompanies this paper as data.

Figure B1 draws both directions of the β Pictoris control, over the same 52 of the 56 crossings, the remainder retaining no control ring. In its second panel the 10 β Pictoris carbon-monoxide crossings land 0.1–0.5 km s^{-1} from CO in the star's own frame, while the 2 unattributed crossings of the same star have no masked transition anywhere near them – the nearest lies 3765–3870 km s^{-1} away – and the methanol components they do sit close to are excluded as their source in Appendix C.

C. THE LINE MASK

A crossing that falls on a known molecular transition has an astrophysical explanation to hand, and the mask exists to say which crossings those are, reporting the frequency coincidence and the judgement about the coinciding species separately.

The list, and when it was made. The mask holds every transition of the 11 species these spectral setups cover – the three CO isotopologues, HCN, HCO^+, CS, CN, SiO, H_2CO, [C I] and hydrogen recombination – that falls inside one of the 46 searched frequency intervals with an upper-state energy below 150 K. That rule is the whole of it, and it admits 96 transitions (Table C1). No strength cut is applied across species: any threshold admitting HCN's strong lines would discard the brightest lines in these fields. Within a species the main hyperfine components are kept and the satellites dropped. The list is a

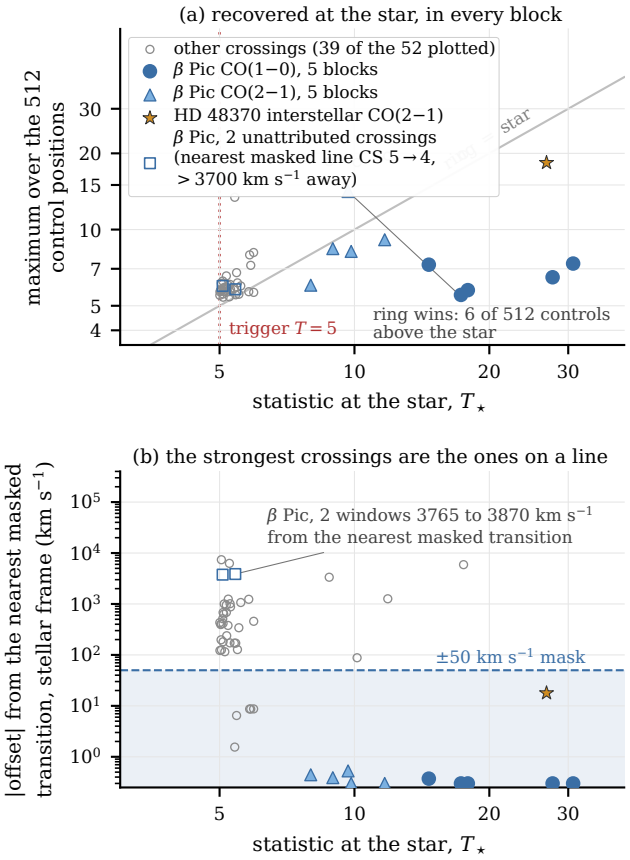


FIG. B1.— The β Pictoris positive control, in both directions (§4.4). *(a)* The spatial screen: the stellar statistic against the maximum over the window's 512 control positions, filled for the blocks recovering β Pictoris CO. *(b)* Velocity offset from the nearest masked transition *in the star's own frame*, the adopted $\pm 50\,\mathrm{km\,s^{-1}}$ half-width marked; grey, the 39 others, HD 48370 labelled.

single Splatalogue query (Remijan et al. 2007) of CDMS and JPL (Pickett et al. 1998; Müller et al. 2005) made on 2026 October 6, and the query is dated and repeatable, so the mask itself can be rebuilt.

The species are the ones the observers went to the telescope for, and that is a limitation. The spectral setups are tuned to them: 38 per cent of the 405 Class A windows are centred within $\pm 50\,\mathrm{km\,s^{-1}}$ of a transition of the list and 87 per cent contain one somewhere, against 1–8 per cent for the same windows displaced bodily. The list is therefore set by the science goals of the programmes that took the data and not by anything in this search, and *unattributed* means unattributed against it. A wider list measures what that costs: every transition below 150 K of all 225 species with a published interstellar or circumstellar detection (Pickett et al. 1998; Müller et al. 2005) gives 24 976 transitions inside the searched intervals, covering 98 per cent of the Class A union against 5.0 per cent for the list adopted here, and it attributes every one of the 38 unattributed crossings together with 100 per cent of the control positions' own exceedances, so it empties the category for star and control alike; narrowed to $\pm 1\,\mathrm{km\,s^{-1}}$ it would attribute 6 of the 38 and the unattributed count remains what the control positions predict, 26 against 28.5 expected. The nearest transition of that wider list is reported beside every unattributed crossing in the accompanying data – and at a density where 98 per cent of the band lies within $50\,\mathrm{km\,s^{-1}}$ of something, proximity is not an identification.

What an identification would look like, on the pair that comes closest to one. The 2 unattributed crossings of β Pictoris lie 6.7 and 7.1 MHz above components of the methanol $5_K \rightarrow 4_K$ ladder ($5_{-1}\rightarrow 4_{-1}$ E and $5_{-3}\rightarrow 4_{-3}$ E, $E_\mathrm{u} = 40$ and 98 K, at frequencies on which CDMS and JPL agree), 8.3 and $8.8\,\mathrm{km\,s^{-1}}$ in the star's frame: offsets agreeing with each other to $0.48\,\mathrm{km\,s^{-1}}$, which is the configuration a real line identification produces and which arises by chance in 0.7 per cent of pairs drawn uniformly in the window they were found in. Four measurements decide against it, all in the one window that covers the whole ladder in both blocks, 241.576–242.007 GHz in the stellar frame. The brightest member of the complex, $5_0\rightarrow 4_0\,\mathrm{A^+}$ at 241791.352 MHz, would reach $T_\star \geq 5.7$ at any excitation temperature and would therefore have been the window maximum, where the maximum over the whole channel–drift grid is the crossing itself. The weaker of the two claimed components cannot exceed 0.65 of the stronger at any temperature and is observed at 0.94 of it. Both fitted drifts are 71 and 30 steps of their own grid from the zero a transition at rest requires, each track crossing 35.7 and 15.2 channels inside its own block. And the two are not in one block: the first is in `Xd9668b_X3a90` and the second in `Xd9668b_Xa9df`, 24.4 h apart, and each is absent from the other, where a persistent line would have been recovered at 5.7 and 4.8σ and the statistic at that cell reaches only 2.45 and 2.88 over every drift of the grid (§4.6). A real pair of components would have stood above the trigger in both blocks; one in each and neither in both is what a noise pair gives. Where this system's own molecular gas is seen, in 10 blocks, it registers to $0.53\,\mathrm{km\,s^{-1}}$ of rest.

Attribution is decided in the star's own rest frame, and the sign is not the usual one. A crossing's frequency is carried from the topocentric value to the barycentre using its own block's pointing and mid-time, and then to the stellar rest frame using the star's systemic velocity. That frame, and no other, is where a coincidence is judged, a tolerance applied in the observed frame spending most of its budget on the Earth's motion. The 3 HD 285968 epochs show the transform working in the only way that can be checked: $12\,\mathrm{km\,s^{-1}}$ apart as observed, they agree to $0.03\,\mathrm{km\,s^{-1}}$ in the star's frame. Throughout, $\Delta v_\star = c\,(\nu_\star - \nu_0)/\nu_0$, so a positive offset means the crossing lies *above* the transition in frequency, the opposite sign to the velocity convention of a spectral line.

Every crossing is read in the local standard of rest as well, where interstellar gas is at rest, and that second reading establishes which coincidences are interstellar. Of the 24 stars carrying crossings, the composite CO(1 → 0) survey of Dame & Thaddeus (2022); Dame et al. (2001) covers 11 and holds emission toward 5 of them, while toward every other star, including the 13 outside its footprint, the all-sky Planck CO map (Planck Collaboration et al. 2016) stays below $0.178\,\mathrm{K\,km\,s^{-1}}$. Toward ALMA J153702653−33192492, HD 139084 B, HD 285968 and HD 48370 the CO spectrum is nonzero at the crossing's own LSR velocity, within $3\,\mathrm{km\,s^{-1}}$ of its own peak and at 0.23 K or brighter against a 0.18 K channel, so 6 of the 18 coincidences are interstellar gas and not circumstellar. Two are worth nam-

TABLE C1
The line mask, by species: the number N of transitions, the frequency range they span inside the searched intervals and their upper-state energies.

Species	N	ν_0 (GHz)	E_u (K)
H	5	92.03 – 231.90	–
H_2CO	13	101.33 – 491.97	10–106
$C^{18}O$	3	109.78 – 329.33	5–32
CN	50	113.12 – 680.30	5–114
CO	3	115.27 – 345.80	6–33
CS	3	195.95 – 342.88	24–66
SiO	2	217.10 – 347.33	31–75
^{13}CO	8	220.40 – 330.59	16–32
HCN	7	354.50 – 354.51	43–43
HCO^+	1	356.73 – 356.73	43
[C I]	1	492.16 – 492.16	24

ing. ALMA J153702653−33192492 is a serendipitous field star toward the Lupus clouds, with the clouds far behind it, and its CO(2 → 1) crossing lies 0.5 km s^{-1} from a 6.6 K km s^{-1} cloud line at +4.9 km s^{-1}: a molecular sightline and not a disc, on which the wider chemistry a cloud carries is a real limitation, 1 of the species such a sightline would show reaching one of its crossings within the mask's own half-width, OCS at 15 km s^{-1}. HD 48370 is the case where the two readings part: its coincidence lies −17.8 km s^{-1} from the star's own CO(2 → 1) while in the local standard of rest it sits at +23.9 km s^{-1}, on a catalogued cloud line peaking at +21.8, so the offset follows the interstellar gas and not the star and that window's extended emission is a cloud and not a resolved disc.

Table C1 gives the mask species by species; the hydrogen recombination lines among them carry no tabulated upper-state energy, and are blank in that column.

The coincidences, and the one species the mask leaves out. Of the 56 crossings, 18 fall within ±50 km s^{-1} of a transition of the list in their own star's frame. The list holds no species whose coincidences are reported and not acted on, so those are also the 18 attributions and the two counts are one count. Sulphur monoxide is a catalogued species of these spectral setups and is deliberately not in the list – no debris disc has a reported SO detection, where the second-generation gas is CO, C and O, so masking its 15 transitions would have declared no constraint over bandwidth this survey does constrain – and the three crossings within the same half-width of one of them (CD−57 1054, CP−72 2713 and HD 161868, at −1.7, −12.3 and −48.0 km s^{-1} from transitions of upper-state energy 21–87 K) are therefore searched, tested and reported as unattributed, with the offset printed beside each in the ledger and 2 of them below the rank screen as well. Interstellar sulphur monoxide is a measurement rather than a preference: in the local standard of rest the three offsets are −3.5, −12.8 and −53.4 km s^{-1}, so for one of them the coincidence does not survive the change of frame at all, 2 lie outside the composite survey's footprint on sightlines carrying at most 0.10 K km s^{-1} of integrated CO against the 1.0 K km s^{-1} a detection needs, and the third, HD 161868, is a molecular sightline whose crossing nevertheless sits 53 km s^{-1} from the emission, where this survey's own cloud test allows 3 km s^{-1}. Each of the three windows also covers CO(3 → 2) and in each the maximum over the whole channel–drift grid is the crossing itself, so no CO(3 → 2) feature above the trigger accompanies any of them – and carbon monoxide, not sulphur monoxide, is what a cloud shows first. The reserved hold-out is treated identically, leaving 2 attributed and 6 unattributed there (§4.6.1).

The half-width, and what it costs. A one-channel test is too tight for circumstellar gas, whose emission is shifted by many channels: the width follows from Keplerian motion where these molecules survive, $\lesssim$13 km s^{-1} beyond 10 au, and is ±50 km s^{-1}. At that width the mask removes 2.38 GHz of the 47.7 GHz Class A union, 5.0 per cent of it, and 3.71 GHz of 118.1 GHz over both classes together, 3.1 per cent, so the cost falls on the primary experiment, because its windows are the line-tuned ones. Attributing on either frame instead would mask the union of two tubes separated by the star's LSR velocity, 2.88 GHz or 6.0 per cent of the Class A union – 21 per cent more bandwidth for no further attribution, which is the measurement behind deciding in one frame.

The labelled crossings are tested for recurrence too (§4.6). For the 2 not recovered a repeat window's own largest statistic bounds the predicted cell from above and the ledger prints the row as a bound, at $T_{rep} < 4.6$ and $T_{rep} < 4.2$ against the $T_{pers} = 8.4$ and $T_{pers} = 6.2$ real carbon monoxide at the discovery flux density would have given. One of those two attributions rests on a velocity rather than on the recurrence test, the sightline carrying catalogued interstellar emission 0.5 km s^{-1} from the crossing's own LSR velocity, and its repeat blocks show where that gas lies rather than whether it can be recovered: in every one of them between 193 and 496 of the 512 control positions stand above the star, so the gas is bright and it is not at the star.

D. THE FALSE-ALARM EXPECTATION AND THE CONTROL ENSEMBLE

This appendix defines the chance expectation for the unattributed crossings, gives the resampling the observed count is compared through, and states what the control ensemble that calibrates it can and cannot support.

D.1. *The expectation, and what it is conditional on*

One attribution set serves both sides of the comparison, the set of §4.3: the mask of Appendix C within ±50 km s^{-1} in the star's own rest frame, which holds no sulphur monoxide, so the three crossings that fall on SO transitions are unattributed crossings and a control maximum on one of those transitions is an unattributed exceedance.

Each of the 405 Class A windows of the primary census contributes two quantities measured on itself. The fraction of its own 512 control positions that reaches the trigger is the probability that a position in that field does so – no cell count enters it and no form is assumed for the tail – and summed over the windows 36.2 are expected to trigger by chance. How many of those would be attributed is measured on the channels the control maxima land on: 88 of 302 exceedances fall inside the mask, carried to the same star's rest frame through the same two steps and against the same transition list as a cell at the stellar position, so the attribution takes 29 per cent of the expectation where a maximum free to land anywhere on its own grid would take 20 per cent: the 15 line-tuned windows lie wholly inside the mask, and a uniform landing would under-attribute them. The atmospheric screen of Appendix E is then applied to these same positions, on the same radius and in the same topocentric frame

as to the crossings, and takes 10 of the 214 exceedances the mask leaves. That leaves 24.6 expected unattributed Class A crossings against 31 observed, as a sum over rows with nothing carried at an assumed rate. Both sides are read over the whole primary beam §3.1 defines, with each block on its own antenna geometry; §4.3.1 reports what a narrower field does to both of them, where the bound below becomes 22. System by system, each system's expectation formed from its own control positions, 2 of the 61 systems fall below one chance in twenty against 3 expected. A further 15 searched Class A windows lie outside the comparison because the deposit does not tabulate them, together with the two unattributed crossings toward HD 14055 they hold, and enter neither side.

A count is compared with the distribution of counts chance delivers, and not with a range on its own mean. A rate is drawn from 20 000 resamplings – each window's rate redrawn from its own 512 probes, the attributed share from its own 302 exceedances, and the execution blocks resampled because windows share them and are not independent trials – and the count scatters about that rate. A count of 31 or more then arises with probability 0.13 at the control positions alone, a probability conditional on the control-position model set out here and not on a null distribution established independently of it. The same resampling bounds what the comparison could have missed: added to each resampled rate, 17 further crossings still leave a count as low as 31 credible at 0.058, and one more does not, at 0.043 against the 5 per cent level. So 17 is the largest added population this comparison could not have rejected against the control positions' own expectation, the uncertainty in 24.6 carried inside it by the same resampling; it counts crossings and not transmitters, and several crossings could come from one source.

The one assumption the searched data cannot test is that a star behaves like a control position in its own field. In the census the star carries 1.104 times as many above-threshold cells as a control position of the same field, measured in the band immediately below the trigger, $4 \leq T < 5$, with 1.046–1.161 allowed at 95 per cent when the execution blocks are resampled; no cell under test enters it, a band count being the difference of two cumulative counts. Calibrating the stellar rate against that ratio rather than against the control positions' own, as a sensitivity test, raises the expectation to 27.1 and the probability to 0.26 and lowers the bound to 15: the conclusion that there is no significant excess is unchanged under it, and the control positions' own rate is adopted as the smaller of the two and the more demanding of the count under test. The requirement runs the other way in any case – the 31 observed would stand above chance at the 5 per cent level only if stars reached the trigger at 0.90 times a control position's rate or less, and every ratio measured here is an excess instead. The reserved blocks measure the same quantity where the search did not reach: their stars trigger in 8 of 62 fine-channel windows against 7.5 predicted, a ratio of 1.07 – though every one of those events lies in a block that shares an observing session with a searched one, and the 28 reserved blocks that share none hold no trigger events between them, so they neither confirm the ratio nor contradict it (Appendix B.1).

Near a band edge the exceedance rate is ×2.464 the interior rate, pooled – and the excess is one object: the window along the sightline to HD 48370 contributes 59 per cent of every near-edge exceedance in the survey, without it the ratio is ×1.020, and on a held-out half it is ×1.181 ($p = 0.24$). No edge correction is therefore applied to any limit.

D.2. *Execution-block quality*

The comparison assumes each block's control ensemble behaves like noise, and one block makes that false: every one of its 512 probes exceeds the trigger, the smallest at 6.6. A criterion is therefore applied to every block and not to the one that prompted it, built from the median of a window's own control ensemble, which separates a field with a source in it from a field that is no null at all: a block is judged on that median divided by the median over the windows of its own class, since a maximum over a larger channel×drift grid sits higher, and the same statistic flags 5 of the 1663 windows individually at twice the survey median, 5.5 against 2.74. *Neither form was fixed in advance*: the condition was noticed in one block and the criterion written afterwards, so it is post hoc, and what makes it trustworthy is how little the position of the line matters – the failing block sits at 4.1 and the next most extreme at 1.12, a factor 3.7 apart, and over 41 thresholds spanning that interval the same single block is selected every time. Of the 407 primary-census blocks, 1 fails; its 4 windows are all coarse-channel and hold the survey's entire Class B crossing population, 4 of the 56 threshold crossings, which appear in the crossing ledger with their block named and enter no count elsewhere.

D.3. *What the ensemble can resolve, and in what unit*

Ranking the star against 512 control positions of the same field decides which windows deserve examination, but $1/(512+1) = 1.95 \times 10^{-3}$ is the *resolution* of the ensemble – the smallest rank it can express – and not the probability that noise produced the window. A survey of 1663 windows would need a rank 65 times finer, and correlation makes it coarser still: controls inside the synthesised beam are not independent, so an annulus resolves no better than 0.023–0.032 however many probes are drawn into it. That floor is the design's central limitation: no single window here can be significant on the rank alone, which is why confirmation requires independent recurrence. The rank distribution is weak evidence in either direction for the same reason, a shift smaller than about 0.042 being invisible in a sample of this size.

The searched cells counted in §3.2 are not independent trials either: channel smoothing and a drift grid stepped twice per channel correlate neighbouring ones, which alone would leave 0.19 of them independent, while fitting each window's effective count against its own control ensemble returns 0.81 and those ensembles' maxima sit at 0.995 of the prediction for the maximum of the full searched count treated as independent and Gaussian. Nothing turns on which is right, because the expectation above is summed from measured rates and not from a cell count: by the cell-count route the number reaching the trigger is 36.6 against 36.2, 3 per cent away. A Gaussian tail over every searched cell predicts 37.9 stellar exceedances against the 56 found; the attribution, the failing block, the archival tail and the atmospheric screen account for the difference.

The widened recurrence window of §4.6 is counted in the unit it is measured in. A window covering more than one crossing is one window and one measurement, since the statistic read is that window's own maximum and does not depend on which crossing it is referred to: the 328 pairings of a crossing with a covering window are 212 distinct windows, 50 covering more than one. Of those, 17 hold a statistic above the trigger against 11.3 expected at $P = 0.05$, or 10.4 at $P = 0.03$ when each window's own control positions supply the rate in place of its cell count; the larger probability is quoted. 10 are toward HD 14055 and ALMA J153702653-33192492 alone, and over the other 161 windows the comparison is 7 against 4.8 at $P = 0.19$, so what departure there is is concentrated and not general.

D.4. *The registration of the predicted cell*

The recurrence test of §4.6 reads one channel at one drift rate, and the cell is registered between two epochs to 0.75 of a channel in every window – half a channel for the prediction, half for the discovery channel it is referred to, and smaller terms for the drift, the systemic velocity and the ephemeris – while the frame correction, the one term that is large in channels rather than small, moves it by up to 116 channels, and the stellar-frame centroid of a real recurring line sits a median $0.05\,\mathrm{km\,s^{-1}}$ from the predicted cell over 61 block readings of the line-attributed recoveries. Each reading was therefore re-formed from the worst cell within $0.45\,\mathrm{km\,s^{-1}}$ of the predicted one with the threshold raised to 5.30 so that the wider reading keeps the single cell's own false-alarm probability, it was pinned at every anchor across the span the discovery track covers, which moves it by at most $16.9\,\mathrm{km\,s^{-1}}$, and the bar itself was reset to 4.3 to hold the chance of one false confirmation across all 444 readings at 1 per cent – and no disposition rests on any of the three, each crossing's own tolerance, cell count, threshold and resulting bound accompanying this paper as data. What the recoveries establish about the registration of a single cell rests on the 3 readings of HD 285968 alone, whose line is confined to one channel; the rest test the frame correction only to the width of a resolved line.

D.5. *From the widened window's exceedances to the pairs*

Two counts of §4.6 are counts of different objects. The 29 above-trigger readings are window readings, one for each pairing of an unattributed crossing with a window that covers its frequency in another block – 328 such pairings over the 37 crossings a later block covers – and they fall in 17 windows of 15 blocks, so a pair whose two members are both unattributed crossings is read twice: 11 of the readings are such a second reading, representing 18 distinct pairs, of which 2 lie outside the interval the pair test admits – the median covering window's half-width, by up to $2105\,\mathrm{km\,s^{-1}}$ – and 16 remain, which is the set Table 5 is formed over; Table D1 gives the accounting crossing by crossing. The observatory's own line-of-sight acceleration is removed from both fitted drifts before they are compared, at most $40\,\mathrm{Hz\,s^{-1}}$ here. Of the 7 pairs whose rates the grid resolves, every one fails a drift changing at a constant rate, by 3.6 standard deviations in the mildest case; the offset $\frac{1}{2}(\dot\nu_1+\dot\nu_2)\Delta t$ such a trajectory predicts has no freedom left once both rates and the interval are fixed, and the one pair the searched grid itself separates is HD 14055's, whose crossings lie $717\,\mathrm{km\,s^{-1}}$ apart in blocks 2.2 hours apart, a single carrier having had to drift at 25 times the grid's ceiling between them. Nothing measured here bounds the curvature in between, the ceiling bounding the line-of-sight component of a carrier's acceleration and not how fast it can turn over; the velocity excursion *is* bounded, to within 1.7 times the circular amplitude over every eccentricity and orientation, which is why the widened window of §4.6 can be set against that excursion and a reversal time cannot. The control row is a line that does not drift: the CO(2–1) line of HD 285968 is recovered at one stellar-frame frequency 5.0 days apart and 0.6 standard deviations from the prediction, so the comparison resolves a mean drift to $103\,\mathrm{Hz\,s^{-1}}$ and no finer at any baseline. For the 12 pairs it cannot separate – 6 of them resolved and failed by the constant-acceleration comparison – nothing measurable tells one carrier from two.

TABLE D1
The 29 above-trigger readings of the widened recurrence window, taken to the pairs of Table 5; each block of rows is the one above with one effect removed.

Covering windows read, over the 37 unattributed crossings a later block covers	328
... holding a statistic above the trigger, in 17 windows of 15 blocks	29
Distinct crossing pairs those readings represent	18
... readings that are the second reading of such a pair	11
Pairs separated by more than $\pm 833\,\mathrm{km\,s^{-1}}$ in the star's frame, to $2105\,\mathrm{km\,s^{-1}}$	2
Pairs inside that interval	16
Pairs with at least one unattributed member	16
... with a fitted drift and a block epoch for both members	13
... separated by the bounded-trajectory criterion	1
... open	12

β Pictoris is the closest case, and it is the one the comparison cannot dismiss. Its two crossings sit at 241.7739 and 241.8594 GHz, $106\,\mathrm{km\,s^{-1}}$ apart in the star's frame, in blocks 24.4 hours apart – both inside the methanol $5_K \rightarrow 4_K$ ladder, which Appendix C excludes as their source – with both fitted drifts negative (-1611 and $-679\,\mathrm{Hz\,s^{-1}}$), so a single carrier would have had to rise in frequency between the epochs and under constant acceleration misses by 132 standard deviations – but the mean acceleration it needs, $1.21\,\mathrm{m\,s^{-2}}$, is inside the $3.6\,\mathrm{m\,s^{-2}}$ ceiling. The pair fixes three numbers, the acceleration at each epoch and the frequency offset between them, and a circular orbit about a star of $1.75\,\mathrm{M_\odot}$ (Crifo et al. 1997) has three free parameters: one orbit reproduces all three at once, at 1.54 d in period, 16.5° in inclination and 0.0314 au in radius, eccentric orbits fit up to e of about 0.28 before the periastron acceleration leaves the ceiling, and the pair is therefore not separable.

D.6. *The drift grid's edges*

The drift trials of a window run from $-\dot\nu_{\max}$ in equal steps of half a native channel across the retained track, and the last trial is the first at or beyond $+\dot\nu_{\max}$, so the interval searched is asymmetric by up to one step and the ceilings themselves, computed window by window from Equation 2, reproduce it to 1×10^{-4} over all 405 Class A windows. Each window's grid is recovered from two quantities that enter no common calculation

TABLE D2
The crossings whose fitted drift lies on an outermost trial of their window's own grid, with that grid. Rates are in the observatory frame.

Star	block	ν (GHz)	$\dot{\nu}_{\rm obs}$ (Hz s^{-1})	trial	$\lvert a_{\rm los}\rvert$ (m s^{-2})
BD+05 1668	`X28a8`	242.195	+3452	2/2	≥ 4.27
HD 14055	`X29ae`	331.156	−3976	0/135	≥ 3.60
HD 14055	`X2c3bb`	345.526	+4184	141/141	≥ 3.63
HD 161868	`X10ca8`	345.634	+4195	127/127	≥ 3.64
TRAPPIST−1	`X259b`	346.245	−4623	0/137	≥ 4.00

– the trial count recorded for it and the channel width and retained span its own crossing record carries – and the two agree on 56 of the 59 crossings that carry a fitted drift, on every one of which the fitted rate is a trial of that grid exactly; 5 lie on an outermost trial (Table D2), toward BD+05 1668, HD 14055, HD 161868, TRAPPIST−1, and are reported as lower bounds. 2 are among the 33 crossings §4.6 bounds; the other 3 carry no amplitude bound, lying in the reserved hold-out, in the quality-failing block and in the one window retaining no discovery spectrum, which is why this table reaches above that section's range. The remaining 2 are in archival-tail windows for which no grid is recorded anywhere.

E. RADIO-FREQUENCY INTERFERENCE SCREENING

Interference is the first alternative explanation for an unattributed crossing, so the screening this survey has and the coverage it lacks are both set out here. Millimetre interferometry is an unusually poor environment for terrestrial interference, though not an empty one: ITU Radio Regulations Article 5 (International Telecommunication Union 2024) carries active allocations across the searched bands, while an interferometer is not a total-power detector, so a signal not in the far field at the phase centre acquires a rapidly varying geometric phase across the array and is suppressed in the correlation rather than imaged. The documented hazard is spaceborne cloud radar in the 94.0–94.1 GHz Earth exploration-satellite allocation (Stephens et al. 2002; Wehr et al. 2023; Emerson 2004): 1 of the 1663 census windows overlap it, carrying 0 threshold crossings and 0 rank-passing events, so no result here rests on it. *The five tests of Table E1 carry none of the signatures interference leaves, and their coverage is uneven.* Each is computed for every searched window of the census, as is the 512-position rank screen, while recurrence reaches only the crossings with a covering repeat block; and because the observatory's own correlator flagging was not retained window by window, *no per-channel interference flag of our own exists, and none can be recovered* – which is also why the five populations differ, each test needing a different quantity to have been recorded. The clustering the crossings do show is the attribution seen from the other side: the one concentration a reader will notice, 15 crossings between 344.2 and 346.4 GHz, is the carbon monoxide tuning, 12 of the 43 unattributed crossings the census and the reserved hold-out together place in a searched window against 11.8 ± 0.9 expected from where the spectral windows were tuned, and that interval holds 0 telluric transitions. The mask must precede the occupancy screen, because 35 of the 109 above-trigger windows sit on a masked transition and a carbon monoxide line falls at nearly the same sky frequency toward every nearby star, which is the coincidence that screen is looking for; applied, it reads 1 against 2.1 ± 1.3. The atmospheric screen is not sharp where it fires – the one crossing it removes sits in a window tuned on the line, 95 per cent of whose channels lie within the same radius of the same transition – so it establishes that this window cannot separate the two and not that the channel is atmospheric; that crossing remains among the 38 unattributed, and the same radius is taken off the control side as one pooled surviving fraction, the larger of the two, so the residual asymmetry runs against the null of Appendix D and not for it. The fifth test bounds the one mechanism that could put a crossing where no line is catalogued: the first local oscillator is not delivered, but each block's own spectral windows pin $\nu_{\rm LO1}$ to a median 0.40 GHz and so place every crossing's image 8–24 GHz away, never inside its own window, while 7 of the 36 survive on frequency alone with the interval taken 1 GHz wider at each end, so it is a bound and not a clearance, and 0 crossings lie where the receivers are double-sideband and no such bound could be drawn. Where a carbon monoxide isotopologue does fall in the image band, 8 crossings lie at the image position against 10.1 expected and neither parent attributes its crossing: each was observed at its own frequency in the same block with nothing at the trigger, the largest amplitude 1.37 and 0.93 times the crossing's own where a leak at the 10 dB image rejection ALMA specifies would need 10 times as much.

One execution block fails these tests, and it fails them as a block and not as a carrier. Its 4 crossings toward BD+05 1668 are the 4 highest statistics in the survey (§4.3.2), and three readings say interference does not account for them: the visibility fit returns a large imaginary part, placing the emission away from the phase centre; the block's 24 sibling executions of the same field, array and spectral setup produce no crossing at all; and of the further windows covering each of its frequencies toward other stars, 0 carries a crossing in the same channel. *None of this substitutes for the ON/OFF cadence a dedicated search uses*: allocation tables cannot exclude harmonics, intermodulation products, unconventional emitters or near-field sources, and archival data without a simultaneous off-source strategy cannot do better. These are statements about the population and not a clearance of any one crossing.

F. PER-SYSTEM SENSITIVITY

The limit Table F1 quotes for a system is that of its best window, so a system whose coverage is wide is not penalised for the shallowest window in it; the window-by-window distribution is the per-window result of §4.1. Each row is one system, a gravitationally bound pair appearing under its primary's name. A window inside the attribution mask constrains nothing (§3.2) and cannot be a best window: a double dagger marks a row keeping less than half its channels, and HIP 10679, whose 2 windows are all inside it, is tabulated with dashes. Beside each limit the table gives that window's centre frequency ν and native channel width $\Delta\nu$, the system's merged coverage $\Delta\nu_{\rm cov}$, the percentage *dec.* by which atmospheric decorrelation leaves the limit optimistic, and whether two blocks of different observing sessions cover

TABLE E1
The five interference tests, each of which could have failed. p is against the same peaks redrawn on their own channel grids.

Test	Population	Result	p
Clustering in sky frequency	51 census crossings with a released peak	11 within $\pm 50\,\mathrm{km\,s^{-1}}$ of a searched transition against 4.8 expected: they cluster on the survey's own lines	0.0010
Crowding on one another	34 unattributed, quality-passing, of the census	no crowding at any of six separations from 1 to 300 MHz, where the 13 attributed crossings crowd at every one	≥ 0.08
Atmospheric transitions	64 crossings of the census and the hold-out, against 570 telluric lines	1 inside the 62.5 MHz screening radius in the topocentric frame, flagged as atmospheric and removed from both sides of the chance comparison while remaining among the unattributed; 1 inside $\pm 50\,\mathrm{km\,s^{-1}}$ against 2.1 expected	–
Fixed intermediate frequency	1401 census windows with a released peak frequency	peak positions within their own windows uniform, median 0.50	–
Cross-target occupancy	3027 extracted windows toward 133 stars, every block set of Table A1 together	569 cross-star coincidences within 5 MHz against 554 ± 13 expected; *below* the null over the 109 above-trigger windows, 4 against 5.2 ± 1.9	0.15
Image sideband	the 36 unattributed crossings lying in a catalogued block	28 have no catalogued transition in the image interval and 4 have one the same block observed with nothing bright enough to leak; for 4 the parent frequency was not observed	–

a common frequency more than a day (d) or more than a year (yr) apart, over the primary census together with the 166 blocks the recurrence test of §4.6 reads, of which 42 lie outside the census – 6 in the archival extension, 22 in the reserved hold-out and 14 in the epoch-extension campaign. The frequency at which a system's best limit falls is set by what the archive holds toward it and not by any property of the star, and for 21 of the 61 systems no sensitivity would have been enough. *The decorrelation column is measured where the delivery allows it and imputed where it does not*: an asterisk marks the 32 systems whose best window lies in a block with no phase record of its own, imputed from the blocks matched to it in baseline, against 27 measured, with a dash for the one system whose blocks offer neither, so the median quoted in §4.1 is a mixture of 4 per cent over the measured and 3 over the imputed. Those blocks are not a random subset – 54 per cent 12 m against 26 per cent of the ones that do expose a record, that array's long baselines being what loses coherence – so each of their windows is matched to measured blocks within a factor of two in baseline length: 1.9 per cent in the median and 21 per cent at the ninetieth percentile, and those limits are conditional on a calibration performance the archive does not report. *Because a system's limit is its best window, a minimum over several estimates could favour windows whose own completeness factor falls below the class average*: drawing each window's factor from the 49 measured ones instead, over 4 000 trials, shifts the per-system median by 5.5 per cent of the adopted 3.03. A dagger marks the 6 of the 60 systems whose own best window was one of the 49 injected into, so that their completeness is measured; the other 54 limits are transferred from the class average and carry instead the dispersion observed between injected windows, $\times 0.80$ to $\times 1.58$, and for the 1 system whose best window lies in a band or channelisation holding no injected window that dispersion is itself transferred.

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TABLE F1

Per-system sensitivity of the carrier search. For each of the 61 systems the fine-channel search covers, the $EIRP_{90}$ of its best window with its own primary-beam, annual-parallax and smearing retentions applied, ordered by the limit. Columns, daggers and dashes are defined above.

System	$EIRP_{90}$ (10^{14} W)	ν (GHz)	$\Delta\nu$ (kHz)	$\Delta\nu_{cov}$ (GHz)	dec. (%)	conf.	System	$EIRP_{90}$ (10^{14} W)	ν (GHz)	$\Delta\nu$ (kHz)	$\Delta\nu_{cov}$ (GHz)	dec. (%)	conf.
LP 876-10	0.87	345.83	488	1.7	4*	d	HR 1010	19	230.53	977	2.8	1	d
AU Mic	1.2	230.56	488	3.5	3*	yr	LHS 1140	19	230.00	488	1.8	0.4*	d
τ Cet	1.2	230.55	488	1.8	0.4*	d	κ CrB	19†	230.57	977	1.7	2*	–
HD 69830	1.2	230.06	488	3.8	1	d	HD 172555	21	340.94	488	5.3	200	–
61 Vir	1.3	345.79	488	1.8	5*	yr	HD 155555C	21†	346.01	488	1.7	43	–
ϵ Eri	1.4	229.98	977	2.3	–	yr	L 836-122	22	346.00	488	1.7	1	–
TRAPPIST-1	1.6	346.58	488	1.7	5	d	CD−57 1054	23	345.99	488	1.7	20	d
β Pic	1.7‡	226.85	122	4.4	2*	yr	CP−72 2713	24	345.12	488	5.8	64	d
AT Mic AB	1.7	346.04	488	1.7	3	–	HD 14055	24	230.11	488	7.2	3	yr
GJ 3379	2.2	229.96	488	1.9	0.4*	d	TWA 7	24	345.99	244	0.87	26*	d
GJ 14	3.1	227.22	488	3.5	19	d	GJ 2006A	24	346.02	488	1.7	1	–
HN Lib	3.9†	229.99	488	1.8	0.4*	d	Wolf 219	26	230.13	488	1.7	2*	d
HD 10647	4.4	345.79	488	6.1	0.8	yr	LP 736-15	28	345.82	977	1.7	4*	–
η Crv	4.5	343.35	488	7.2	27	yr	TYC 2211-1309-1	28	346.00	488	1.7	5	–
HD 202628	5.4	230.51	488	1.8	4*	yr	UCAC3 176-23654	29	346.02	488	1.7	4	–
HD 53143	5.4	230.53	977	1.8	2*	yr	PM J03433+1958	31	230.13	488	1.7	2*	–
HD 207129	6.4	230.57	488	1.8	3*	yr	UCAC2 20312880	32	345.96	488	1.7	11	–
GJ 849	6.8	230.02	488	1.9	0.4*	d	LP 353-51	33	346.02	488	1.7	26	–
TWA 3A	7.6†	230.54	122	0.44	35*	d	Barta 161 12	34	346.02	488	1.7	4*	–
HD 139664	7.8	230.55	488	1.7	3*	–	51 Eri	37	230.53	1953	1.7	11*	–
HD 92945	8.3	355.11	488	3.7	6*	d	HD 23484	37	230.53	977	1.8	0.2	d
TX PsA	8.5	346.03	488	1.7	7	–	HD 139084	54	346.01	488	1.7	3	–
HD 285968	9.1	229.97	488	1.8	0.4*	d	WD 0407+179	84	230.13	488	1.7	2*	–
HD 161868	9.9	227.22	488	7.4	5	d	HD 48370	88†	230.73	122	3.3	0.4*	yr
HD 38858	13	230.50	488	1.8	3*	yr	HD 45184	93	230.53	977	1.7	2	d
HD 110411	13	219.58	244	4.7	2*	yr	ALMA J153702653-33192492	99	345.98	488	14	5*	d
HIP 490	13†	230.53	244	0.86	3*	–	HD 31392	130	230.52	977	1.7	0.9	d
HD 107146	14	355.14	488	3.7	6*	d	HD 89452	260	230.53	977	1.7	0.7	d
LP 476-207	15	346.02	488	1.7	4	–	HD 377	370	230.54	977	1.7	68	d
HD 170773	17	230.13	488	7.4	2*	d	HIP 10679	–	–	–	0.11	–	–
HD 61005	19	230.52	244	19	3*	d							

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